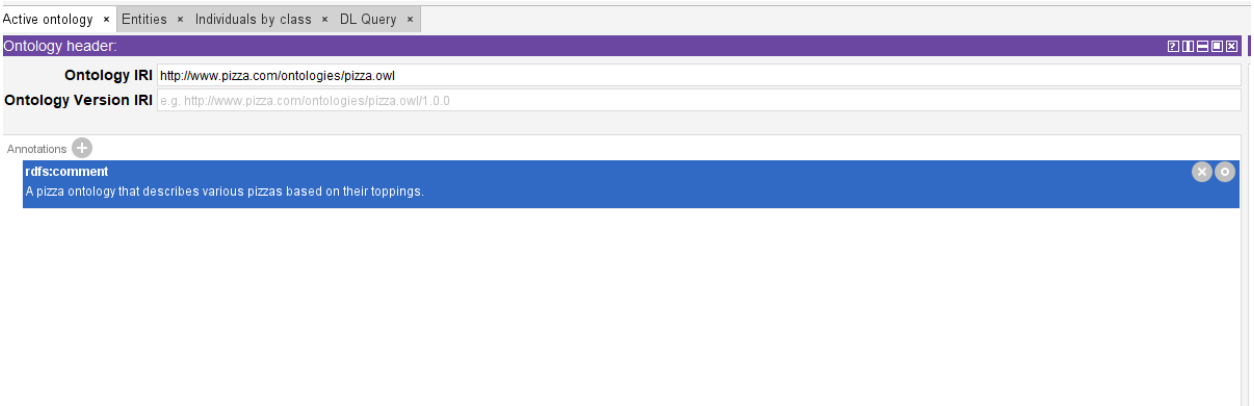
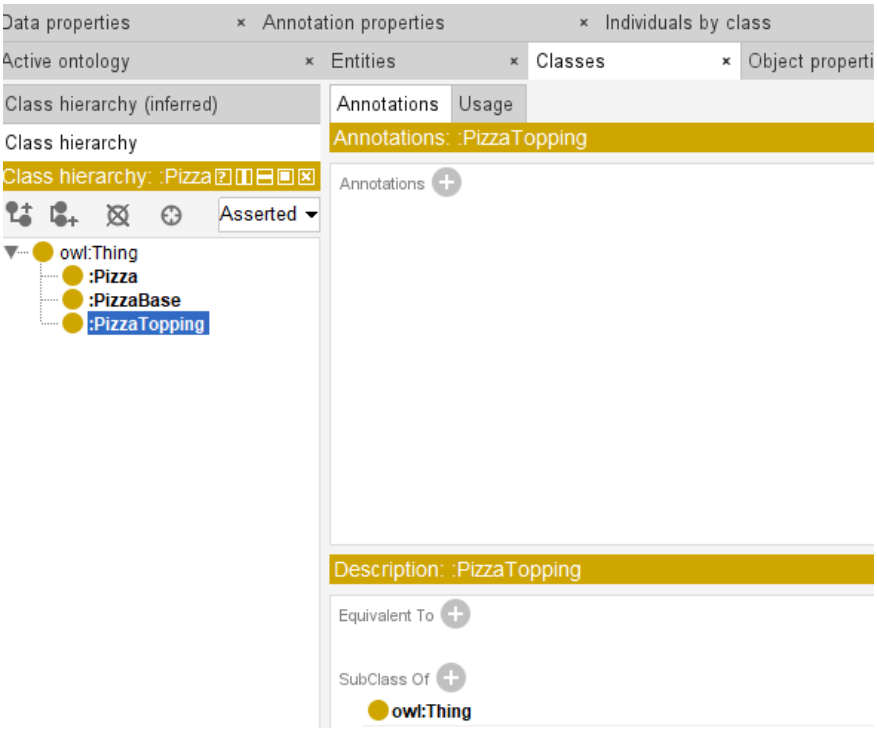
# Team: Sakka Mohamad-Mario, Mirghani Abdelrahman 1241EB

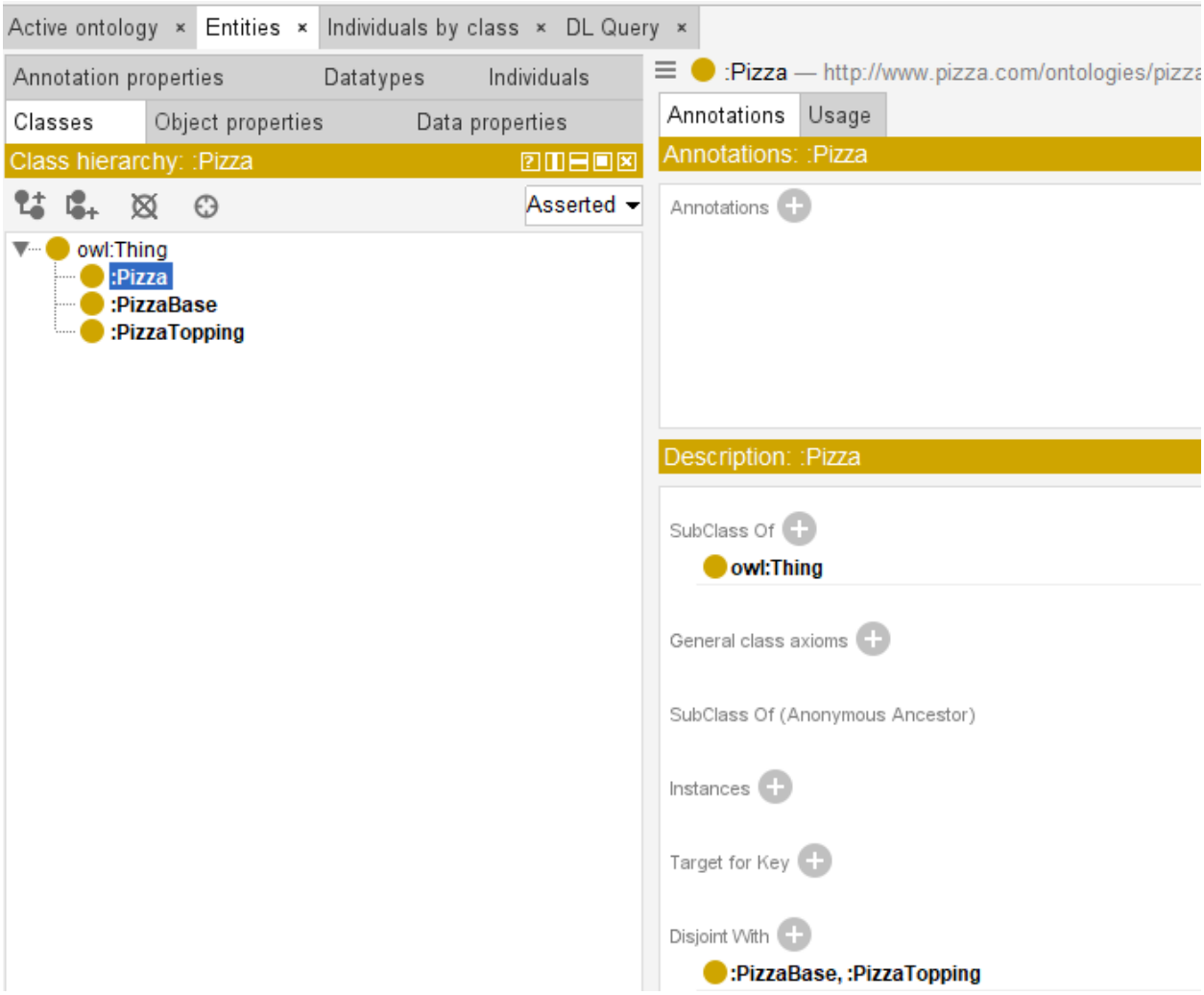
# Exercises 2 & 3



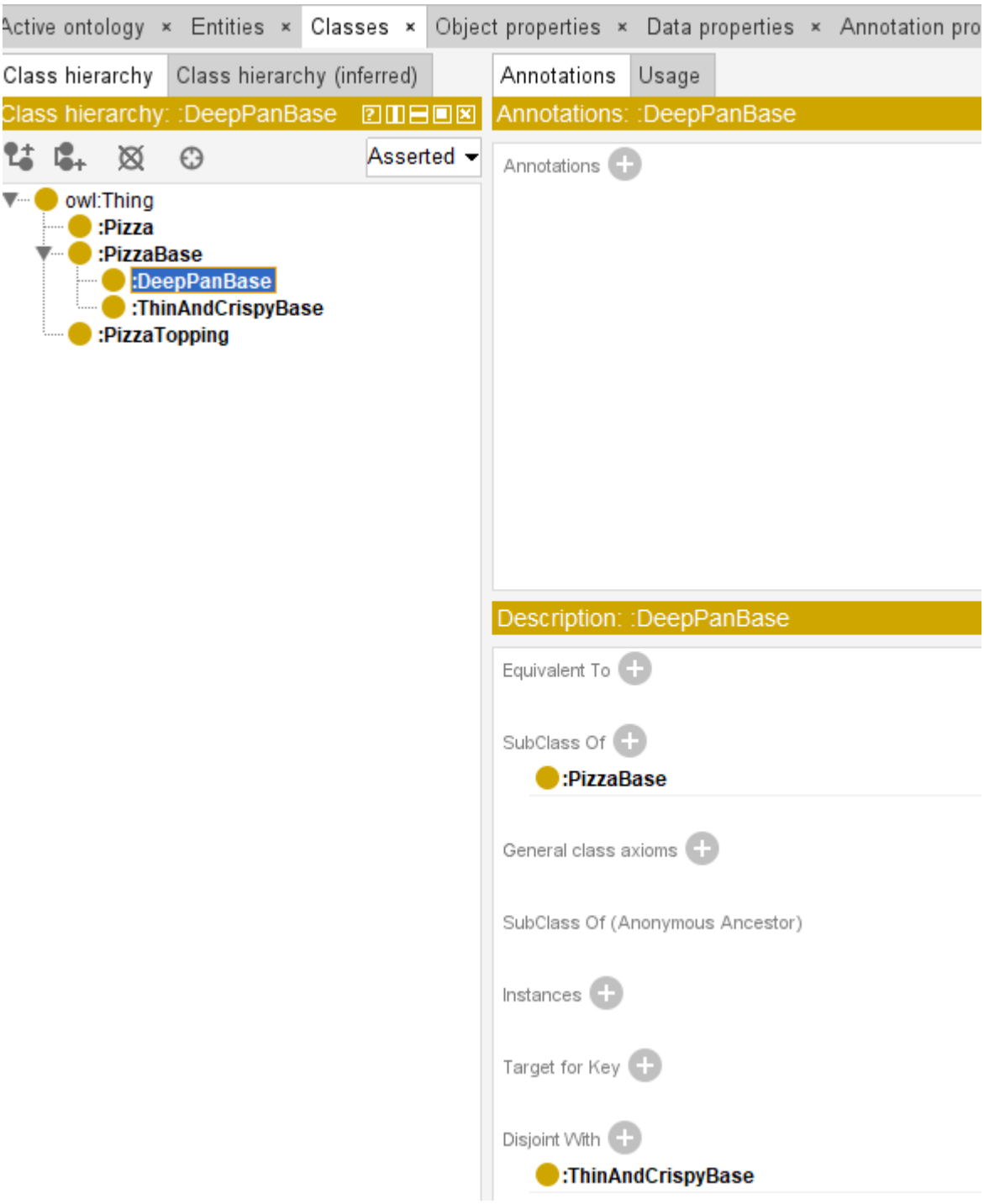

# Exercise 4



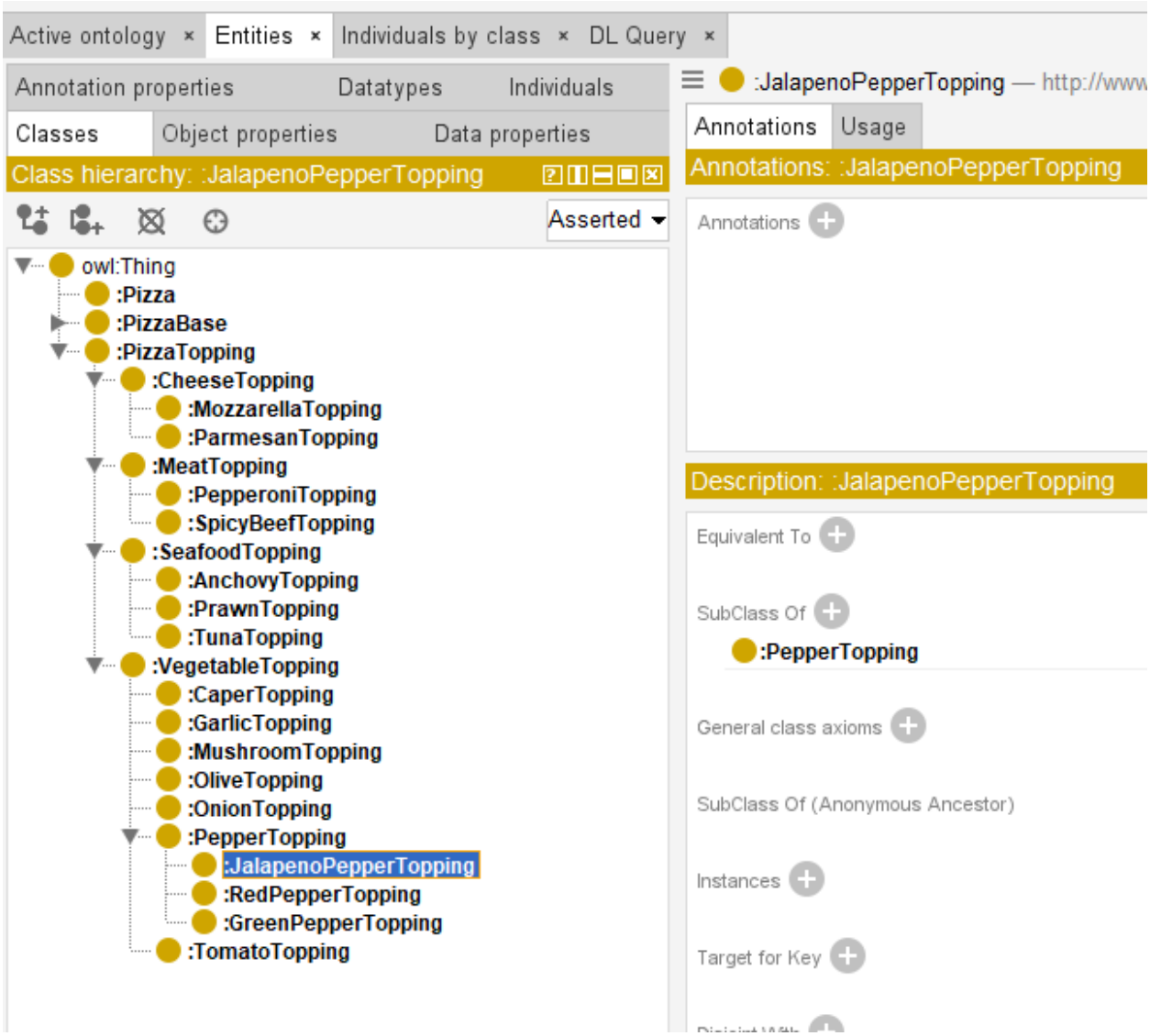
# Exercise 5



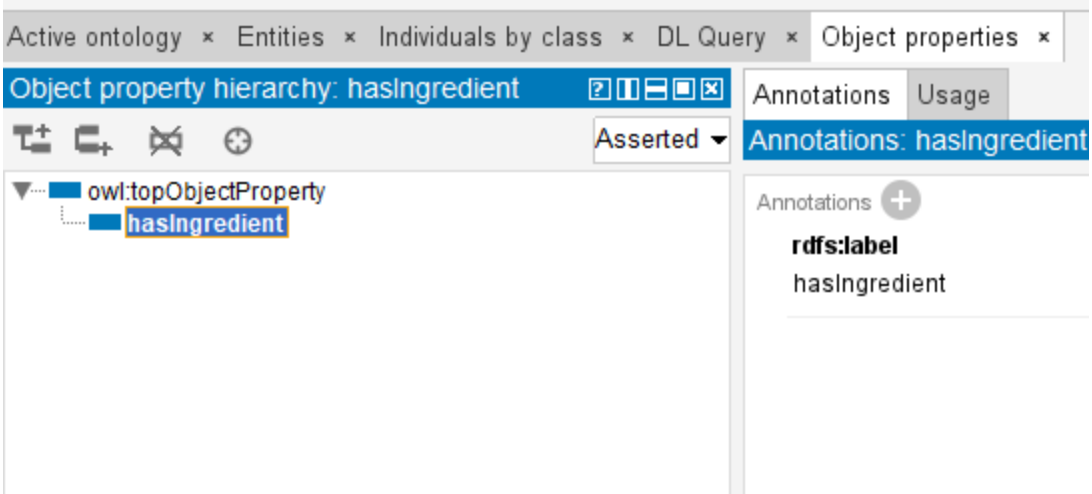

# Exercise 6


Active ontology
Entities
Classes
Object properties
Data properties
Annotation pro
Class hierarchy
Class hierarchy (inferred)
Annotations
Usage
Class hierarchy: :DeepPanBase
Annotations: :DeepPanBase
Asserted
owl:Thing
:Pizza
:PizzaBase
:DeepPanBase
:ThinAndCrispyBase
:PizzaTopping
Annotations
Description: :DeepPanBase
Equivalent To
SubClass Of
:PizzaBase
General class axioms
SubClass Of (Anonymous Ancestor)
Instances
Target for Key
Disjoint With
:ThinAndCrispyBase

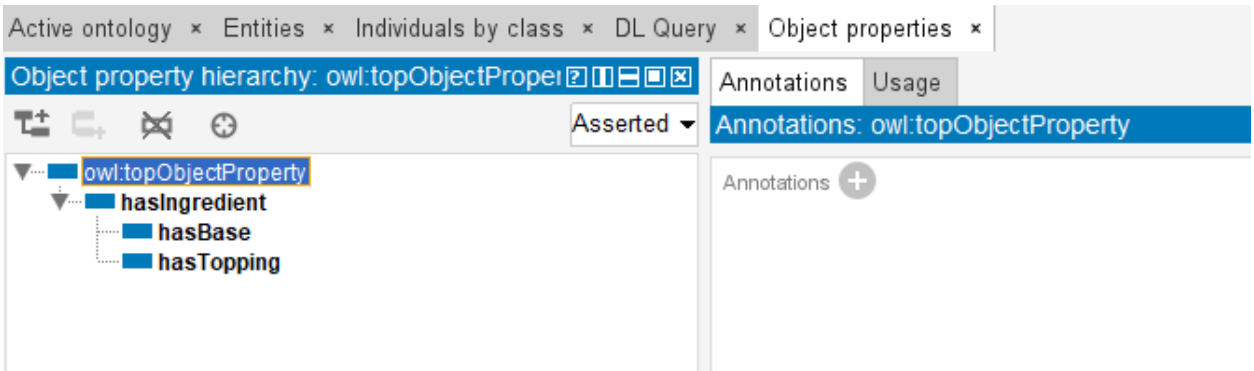
# Exercise 7


Active ontology
Entities
Individuals by class
DL Query
Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Class hierarchy: :JalapenoPepperTopping
Asserted
owl:Thing
:Pizza
:PizzaBase
:PizzaTopping
:CheeseTopping
:MozzarellaTopping
:ParmesanTopping
:MeatTopping
:PepperoniTopping
:SpicyBeefTopping
:SeafoodTopping
:AnchovyTopping
:PrawnTopping
:TunaTopping
:VegetableTopping
:CaperTopping
:GarlicTopping
:MushroomTopping
:OliveTopping
:OnionTopping
:PepperTopping
:JalapenoPepperTopping
:RedPepperTopping
:GreenPepperTopping
:TomatoTopping
:JalapenoPepperTopping — http://www
Annotations
Usage
Annotations: :JalapenoPepperTopping
Annotations
Description: :JalapenoPepperTopping
Equivalent To
SubClass Of
:PepperTopping
General class axioms
SubClass Of (Anonymous Ancestor)
Instances
Target for Key

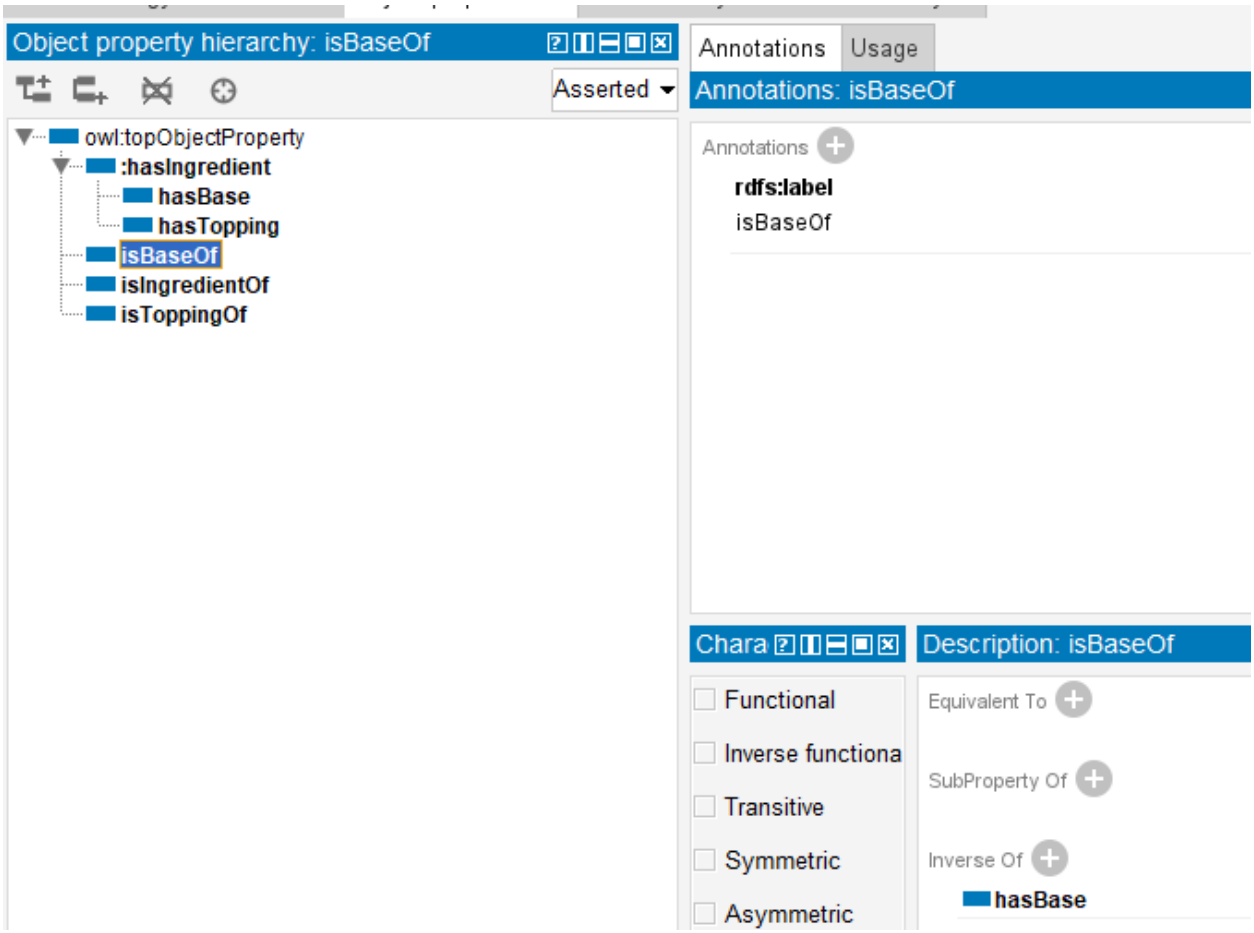
# Exercise 8



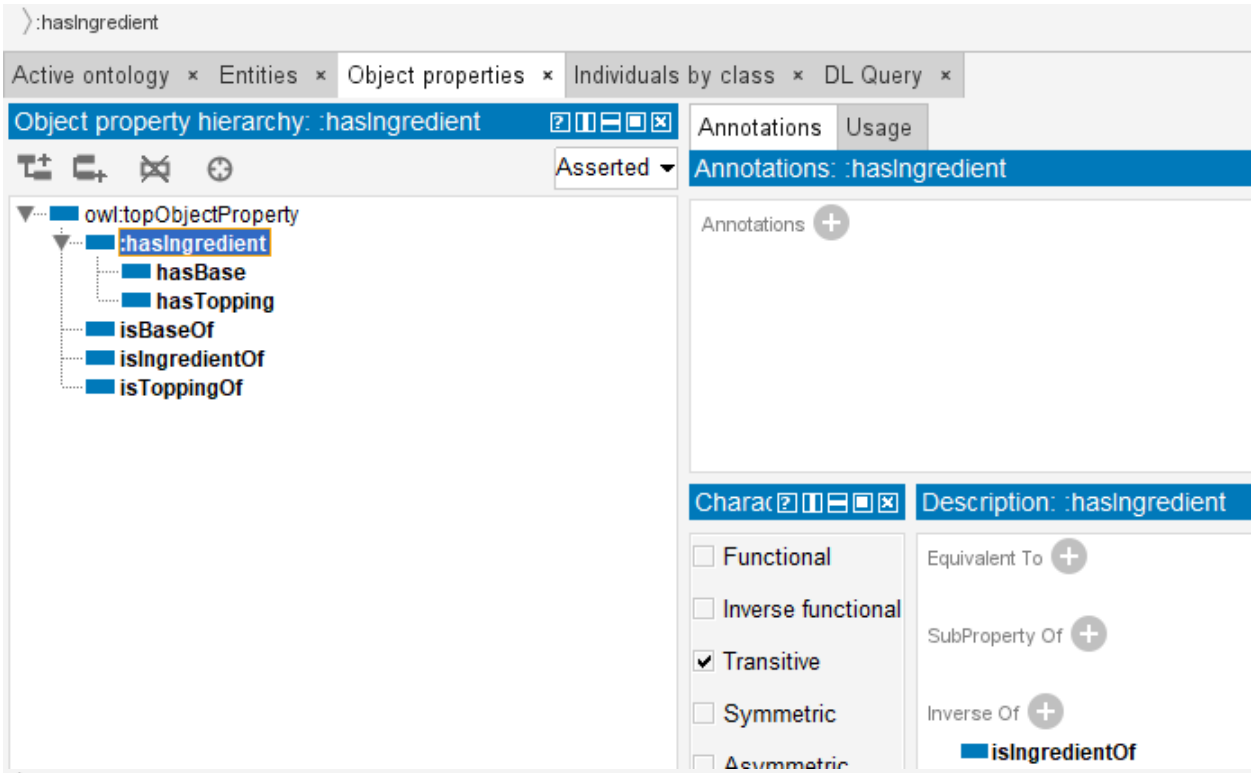

# Exercise 9

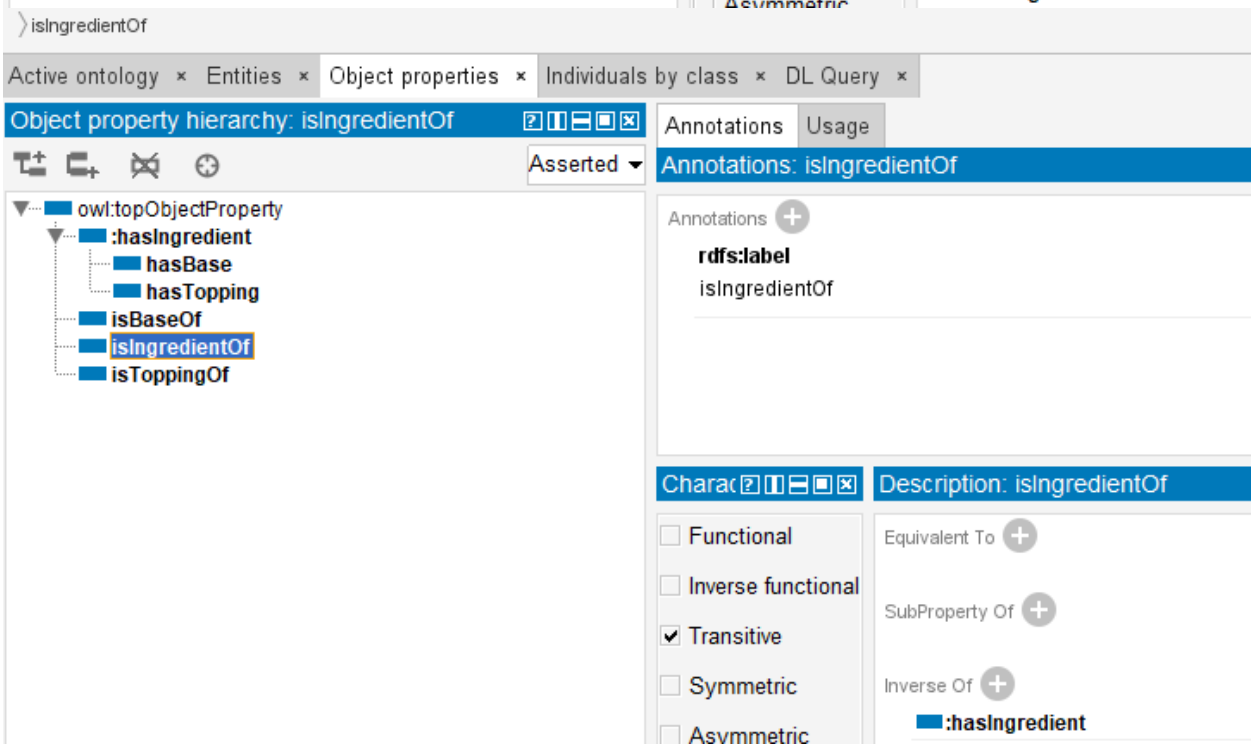



# Exercise 10

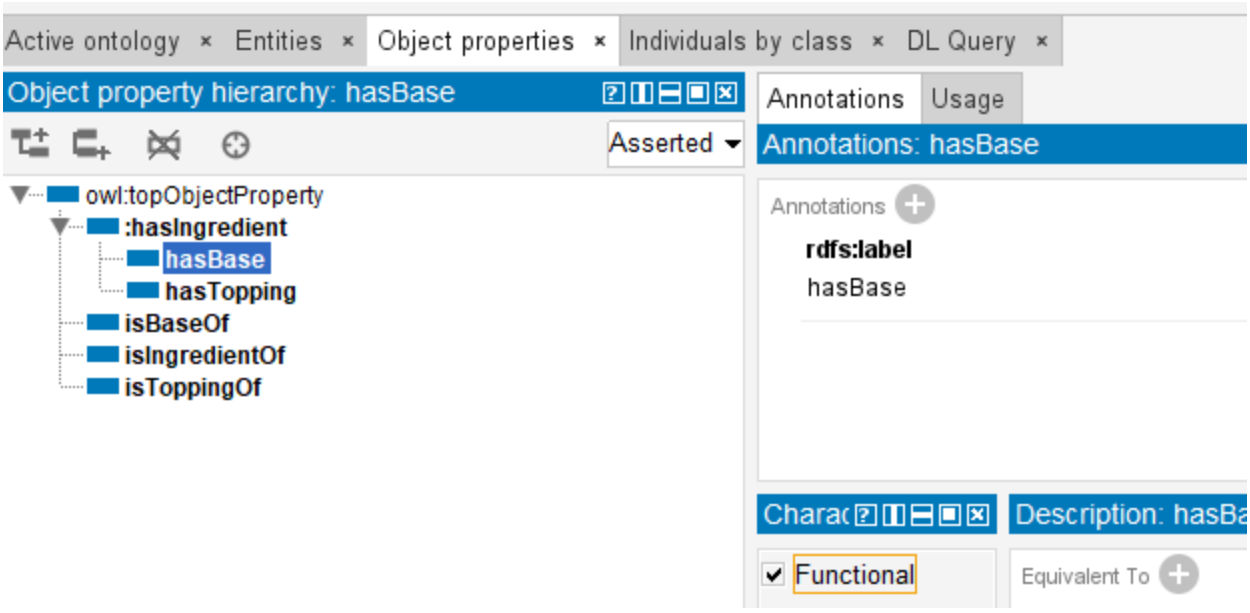

Object property hierarchy: isBaseOf
Asserted
owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
isBaseOf
isIngredientOf
isToppingOf
Annotations
Usage
Annotations: isBaseOf
Annotations
rdfs:label
isBaseOf
Chara
Functional
Inverse functiona
Transitive
Symmetric
Asymmetric
Description: isBaseOf
Equivalent To
SubProperty Of
Inverse Of
hasBase

# Exercise 11

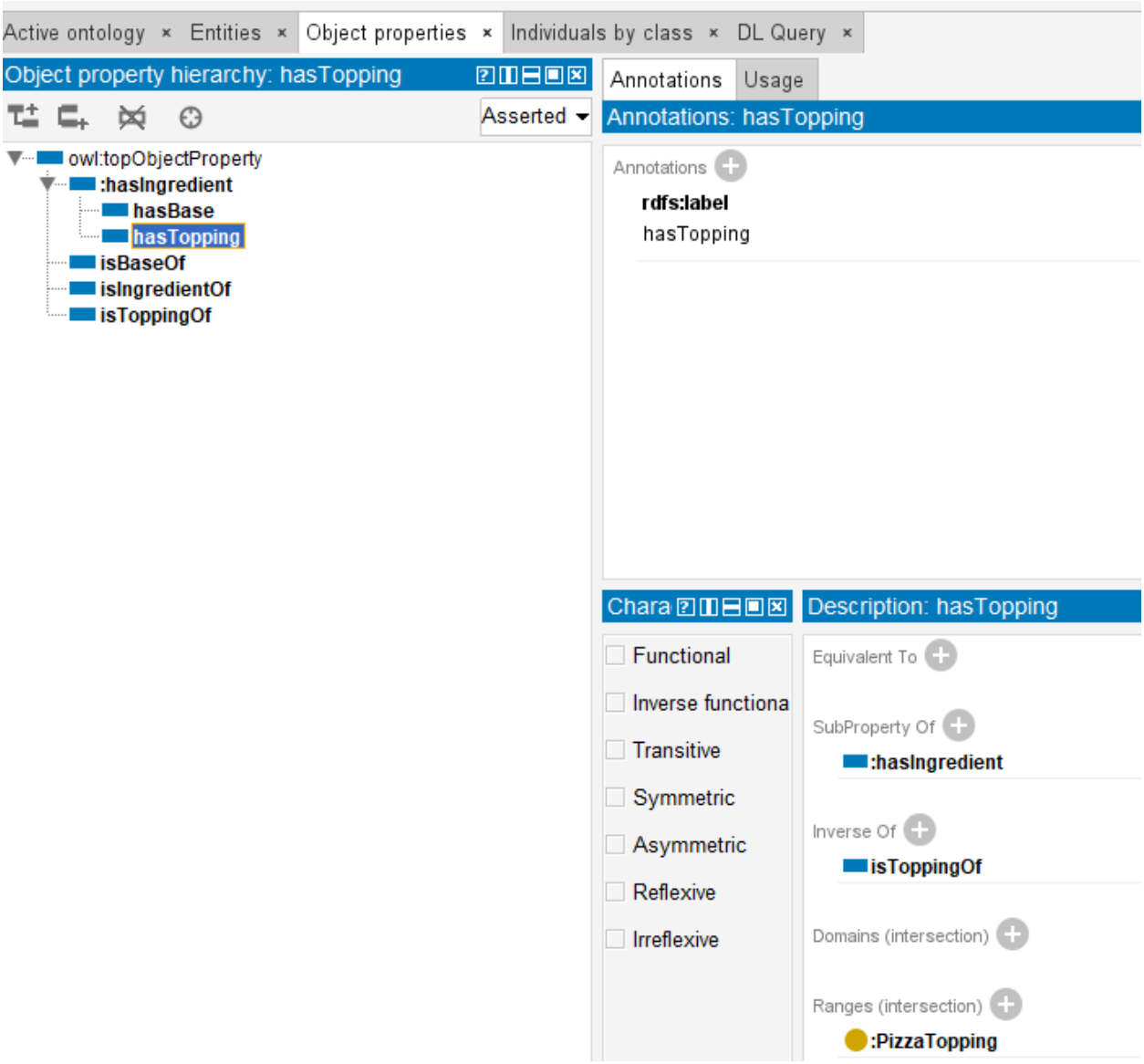

:hasIngredient
Active ontology
Entities
Object properties
Individuals by class
DL Query
Object property hierarchy: :hasIngredient
Asserted
owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
isBaseOf
isIngredientOf
isToppingOf
Annotations
Usage
Annotations: :hasIngredient
Annotations
Charac
Description: :hasIngredient
Functional
Inverse functional
Transitive
Symmetric
Equivalent To
SubProperty Of
Inverse Of
isIngredientOf


isIngredientOf
Active ontology
Entities
Object properties
Individuals by class
DL Query
Object property hierarchy: isIngredientOf
Asserted
owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
isBaseOf
isIngredientOf
isToppingOf
Annotations
Usage
Annotations: isIngredientOf
Annotations
rdfs:label
isIngredientOf
Charac
Description: isIngredientOf
Functional
Inverse functional
Transitive
Symmetric
Asymmetric
Equivalent To
SubProperty Of
Inverse Of
:hasIngredient

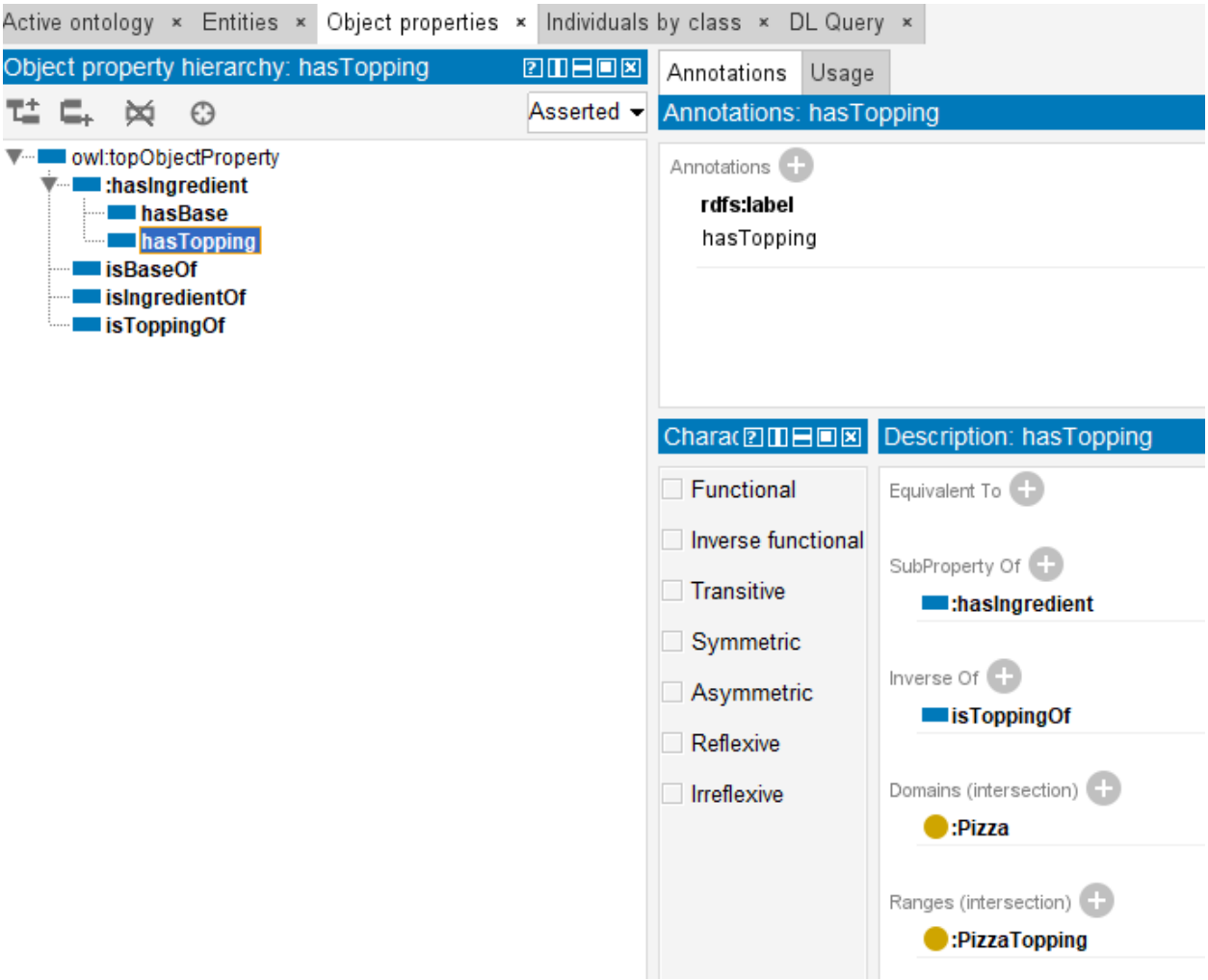
# Exercise 12


Active ontology
Entities
Object properties
Individuals by class
DL Query
Object property hierarchy: hasBase
Asserted
owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
isBaseOf
isIngredientOf
isToppingOf
Annotations
Usage
Annotations: hasBase
Annotations
rdfs:label
hasBase
Charac
Description: hasBa
Functional
Equivalent To

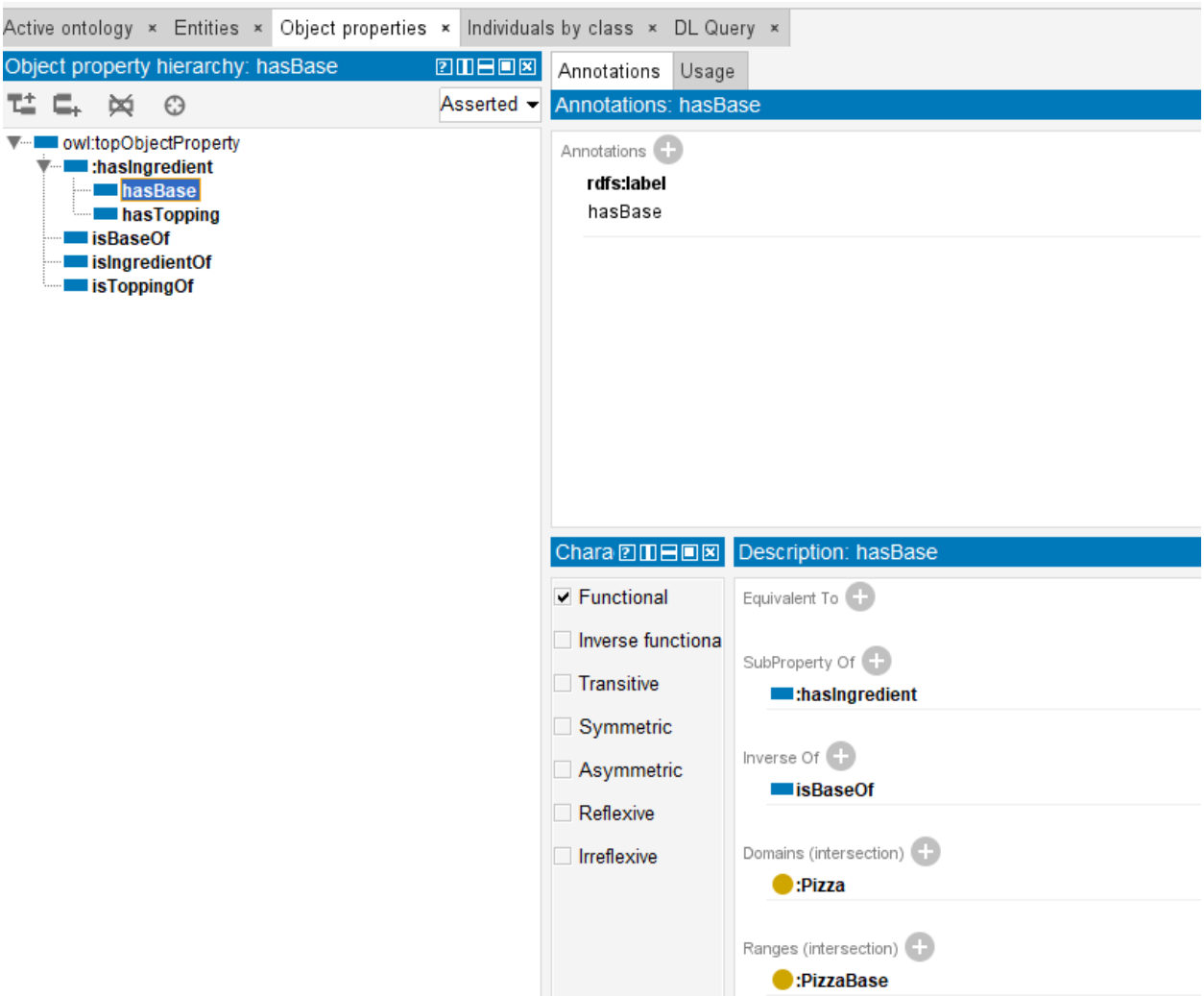
# Exercise 13

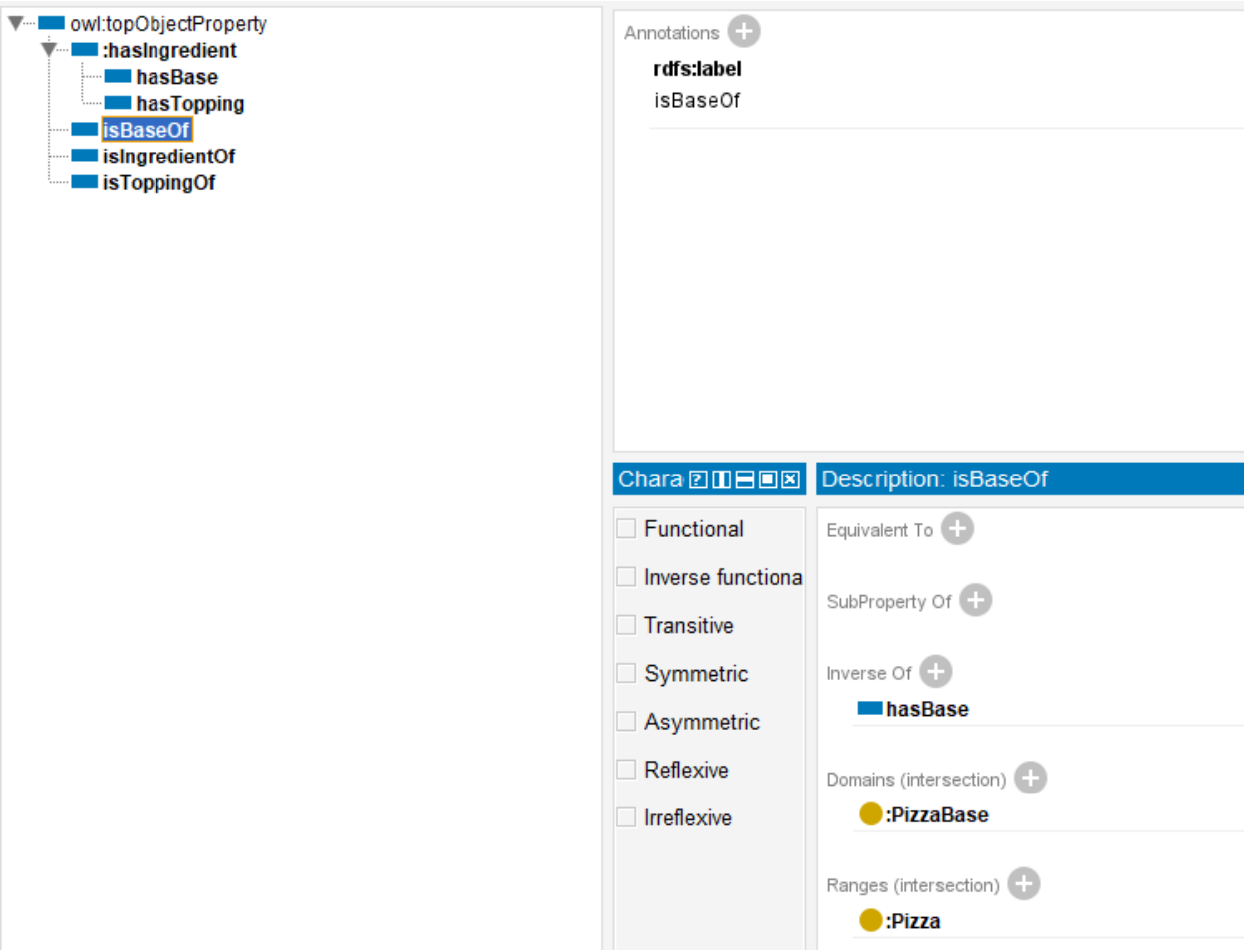

Active ontology
Entities
Object properties
Individuals by class
DL Query
Object property hierarchy: hasTopping
Asserted
owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
isBaseOf
isIngredientOf
isToppingOf
Annotations
Usage
Annotations: hasTopping
Annotations
rdfs:label
hasTopping
Chara
Functional
Inverse functiona
Transitive
Symmetric
Asymmetric
Reflexive
Irreflexive
Description: hasTopping
Equivalent To
SubProperty Of
:hasIngredient
Inverse Of
isToppingOf
Domains (intersection)
Ranges (intersection)
:PizzaTopping

# Exercise 14

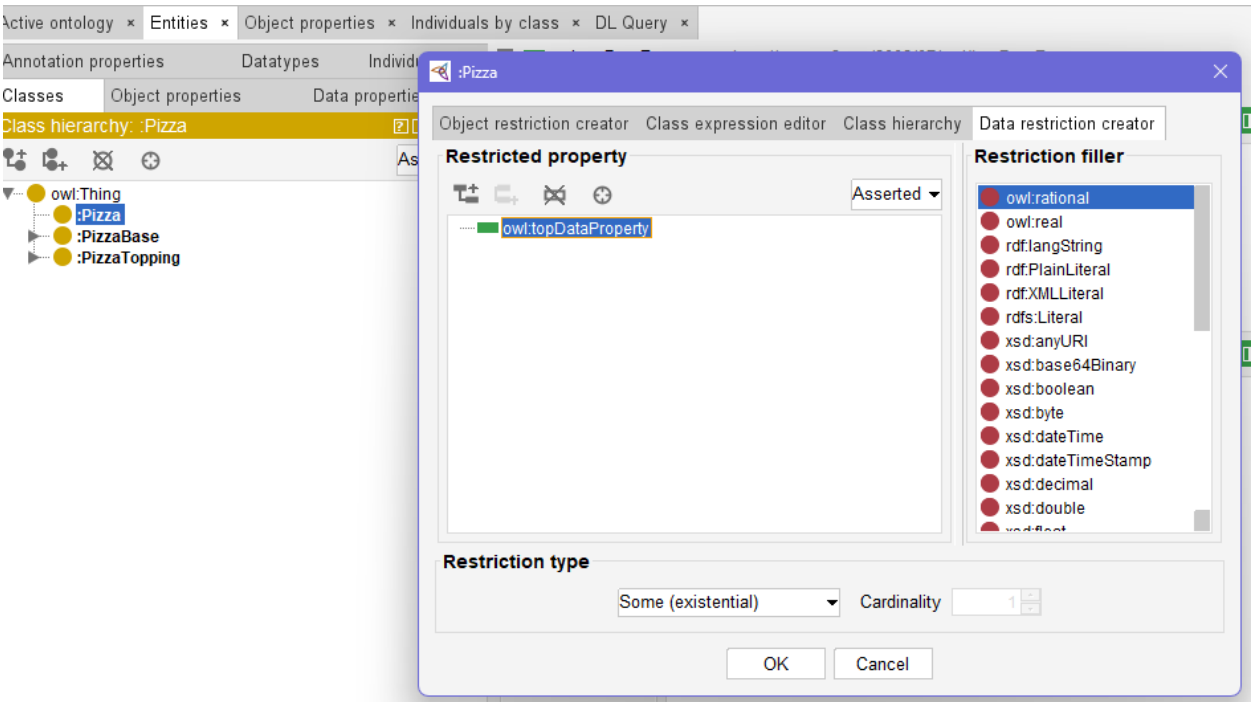

Active ontology
Entities
Object properties
Individuals by class
DL Query
Object property hierarchy: hasTopping
Asserted
owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
isBaseOf
isIngredientOf
isToppingOf
Annotations
Usage
Annotations: hasTopping
Annotations
rdfs:label
hasTopping
Charac
Functional
Inverse functional
Transitive
Symmetric
Asymmetric
Reflexive
Irreflexive
Description: hasTopping
Equivalent To
SubProperty Of
:hasIngredient
Inverse Of
isToppingOf
Domains (intersection)
:Pizza
Ranges (intersection)
:PizzaTopping

# Exercise 15

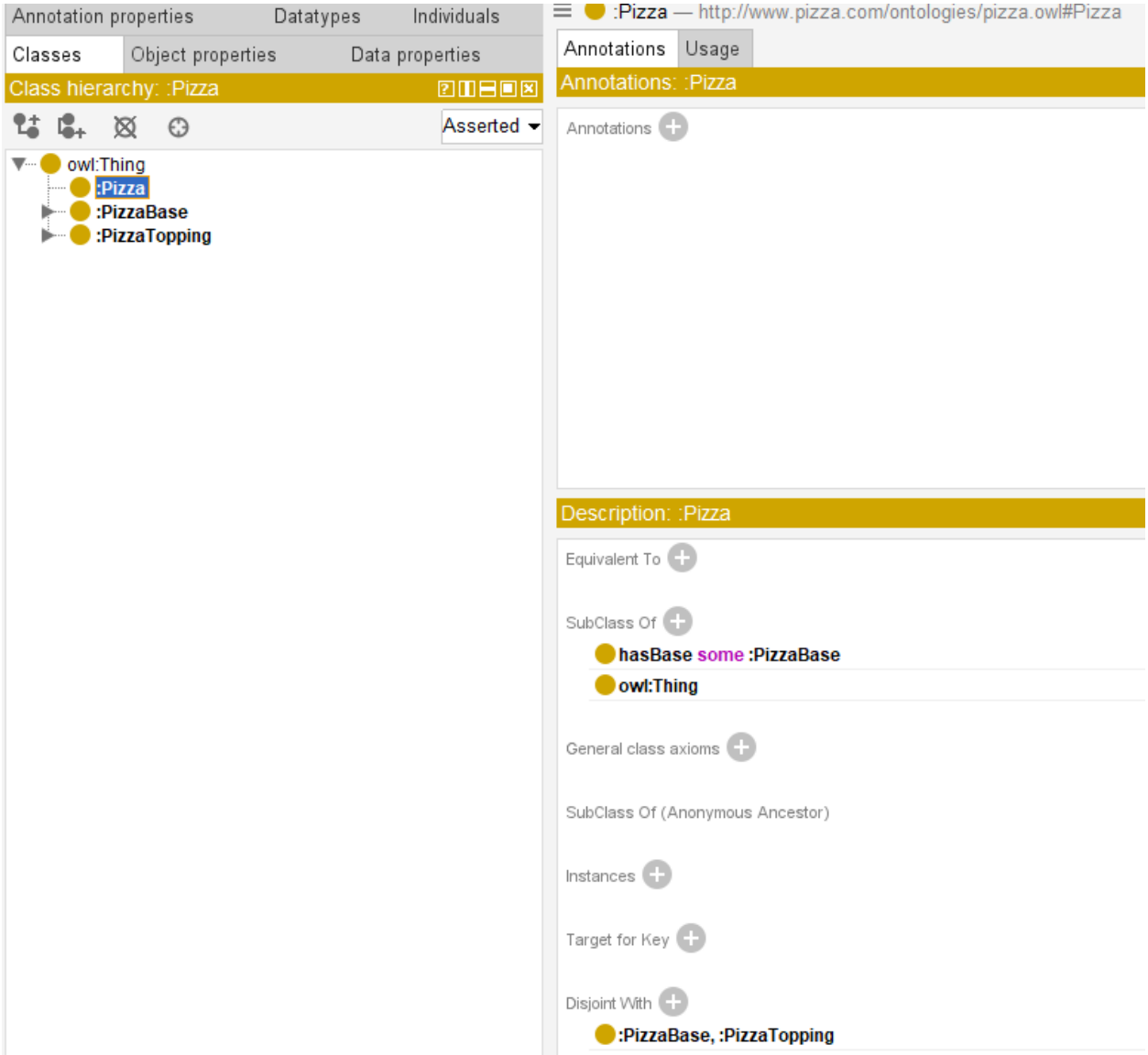

Active ontology
Entities
Object properties
Individuals by class
DL Query
Object property hierarchy: hasBase
Asserted
owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
isBaseOf
isIngredientOf
isToppingOf
Annotations
Usage
Annotations: hasBase
Annotations
rdfs:label
hasBase
Chara
Functional
Inverse functiona
Transitive
Symmetric
Asymmetric
Reflexive
Irreflexive
Description: hasBase
Equivalent To
SubProperty Of
:hasIngredient
Inverse Of
isBaseOf
Domains (intersection)
:Pizza
Ranges (intersection)
:PizzaBase


owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
isBaseOf
isIngredientOf
isToppingOf
Annotations
rdfs:label
isBaseOf
Chara
Description: isBaseOf
Functional
Inverse functiona
Transitive
Symmetric
Asymmetric
Reflexive
Irreflexive
Equivalent To
SubProperty Of
Inverse Of
hasBase
Domains (intersection)
:PizzaBase
Ranges (intersection)
:Pizza

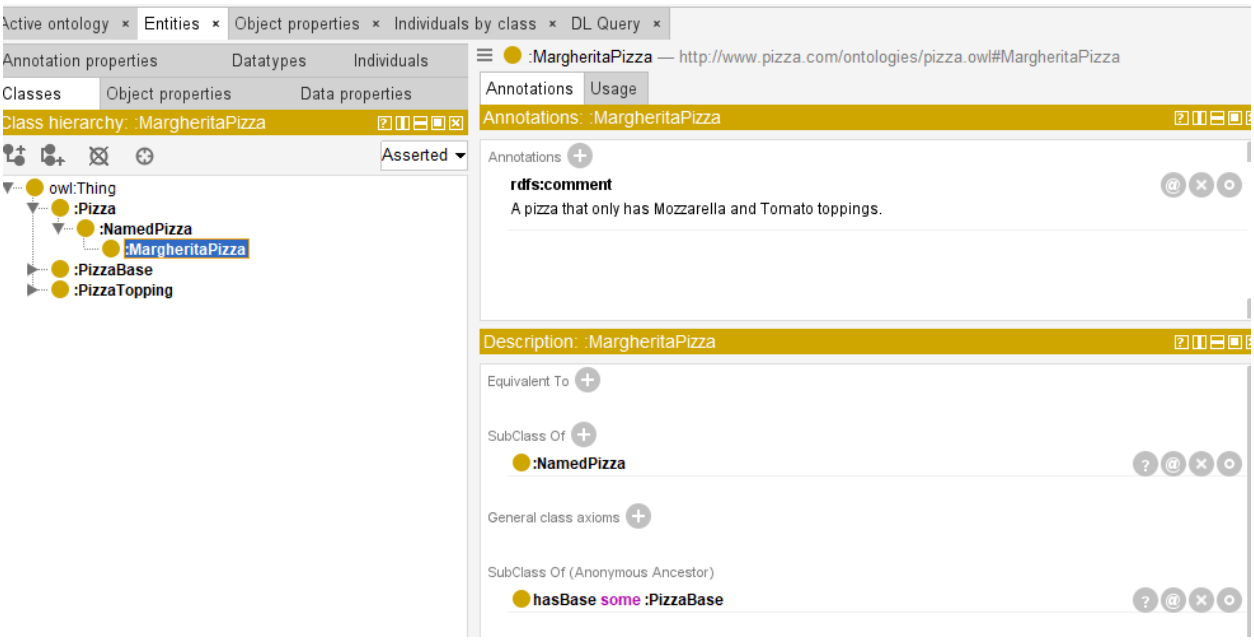
# Exercise 16


Active ontology
Entities
Object properties
Individuals by class
DL Query
Annotation properties
Datatypes
Classes
Object properties
Class hierarchy: :Pizza
owl:Thing
:Pizza
:PizzaBase
:PizzaTopping
:Pizza
Object restriction creator
Class expression editor
Class hierarchy
Data restriction creator
Restricted property
Asserted
owl:topDataProperty
Restriction filler
owl:rational
owl:real
rdf:langString
rdf:PlainLiteral
rdf:XMLLiteral
rdfs:Literal
xsd:anyURI
xsd:base64Binary
xsd:boolean
xsd:byte
xsd:dateTime
xsd:dateTimeStamp
xsd:decimal
xsd:double
Restriction type
Some (existential)
Cardinality
OK
Cancel

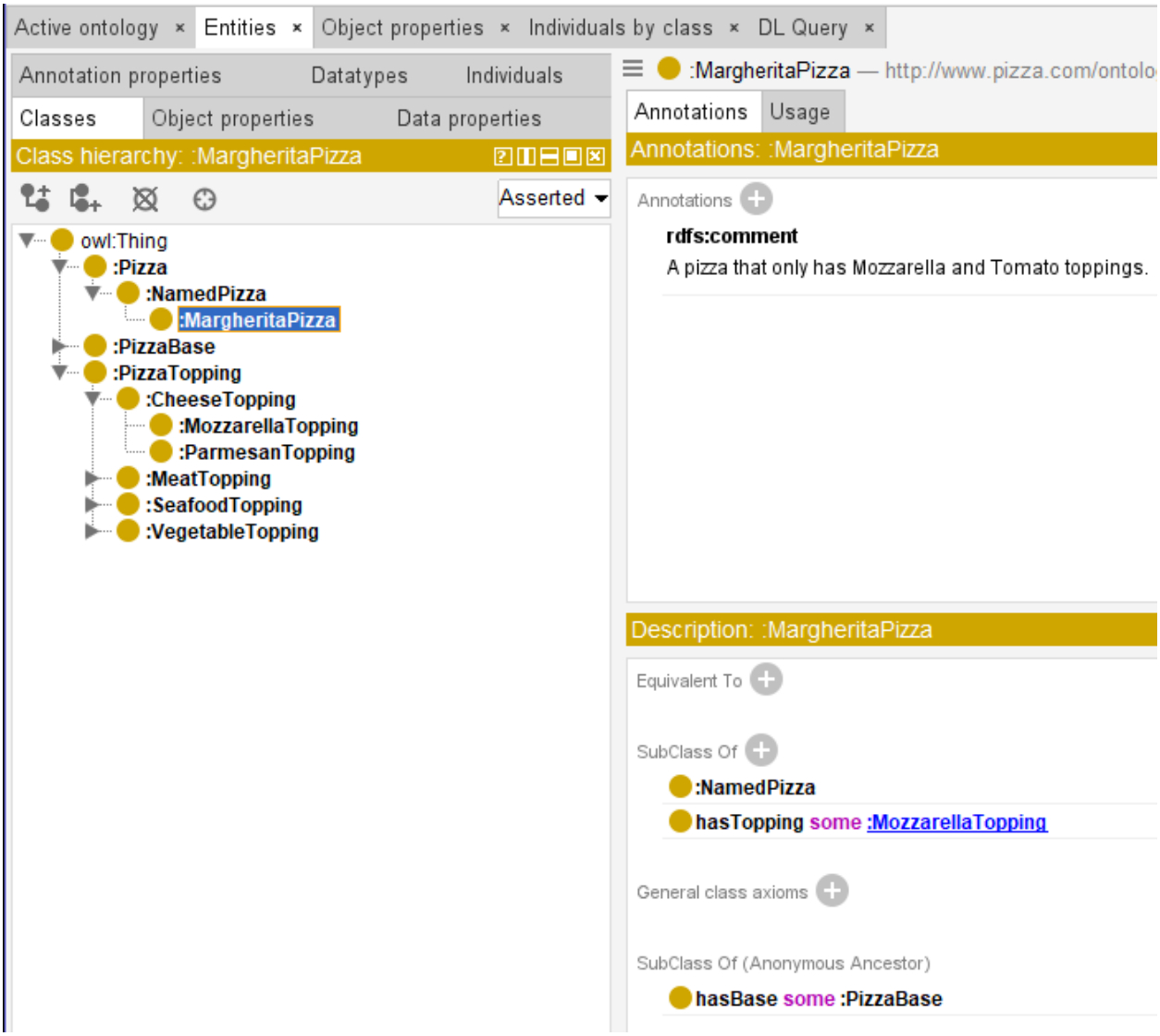
# Exercise 17


Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Class hierarchy: :Pizza
Asserted
owl:Thing
:Pizza
:PizzaBase
:PizzaTopping
:Pizza — http://www.pizza.com/ontologies/pizza.owl#Pizza
Annotations
Usage
Annotations: :Pizza
Annotations
Description: :Pizza
Equivalent To
SubClass Of
hasBase some :PizzaBase
owl:Thing
General class axioms
SubClass Of (Anonymous Ancestor)
Instances
Target for Key
Disjoint With
:PizzaBase, :PizzaTopping

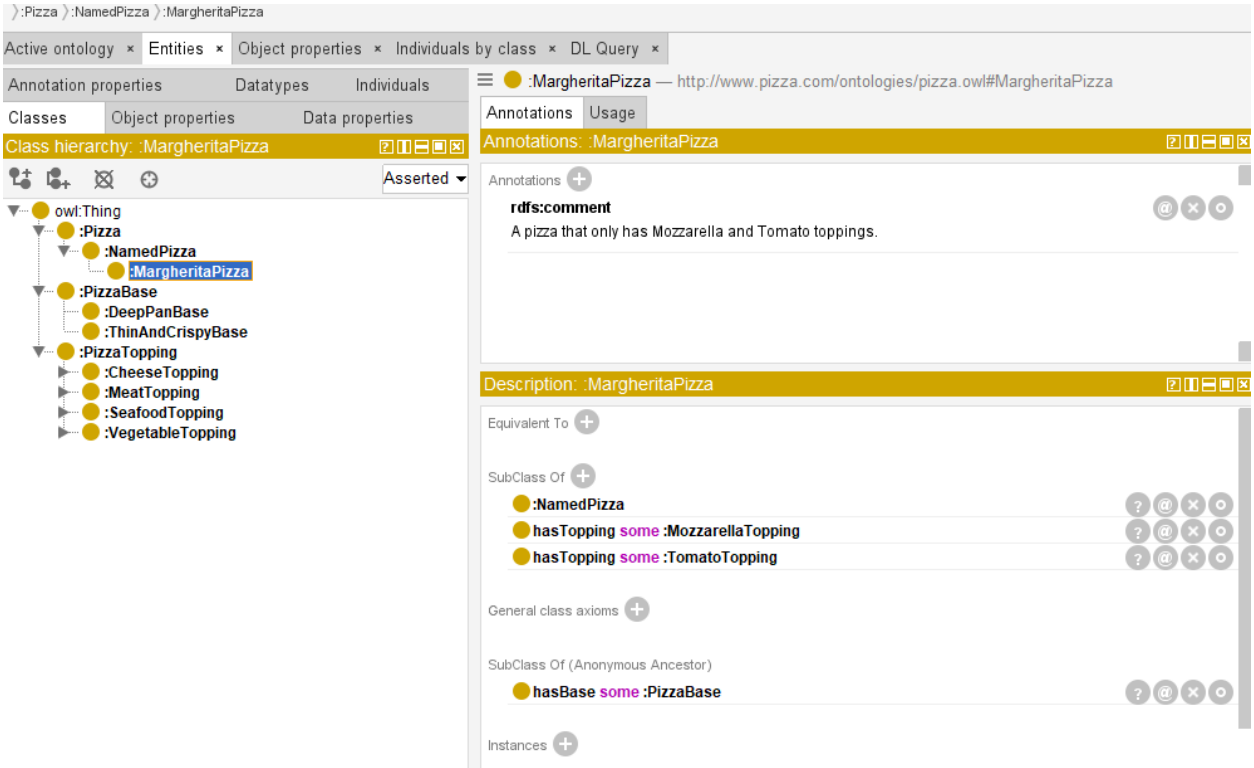
# Exercise 18


Active ontology
Entities
Object properties
Individuals by class
DL Query
Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Class hierarchy: :MargheritaPizza
Asserted
owl:Thing
:Pizza
:NamedPizza
:MargheritaPizza
:PizzaBase
:PizzaTopping
:MargheritaPizza — http://www.pizza.com/ontologies/pizza.owl#MargheritaPizza
Annotations
Usage
Annotations: :MargheritaPizza
Annotations
rdfs:comment
A pizza that only has Mozzarella and Tomato toppings.
Description: :MargheritaPizza
Equivalent To
SubClass Of
:NamedPizza
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

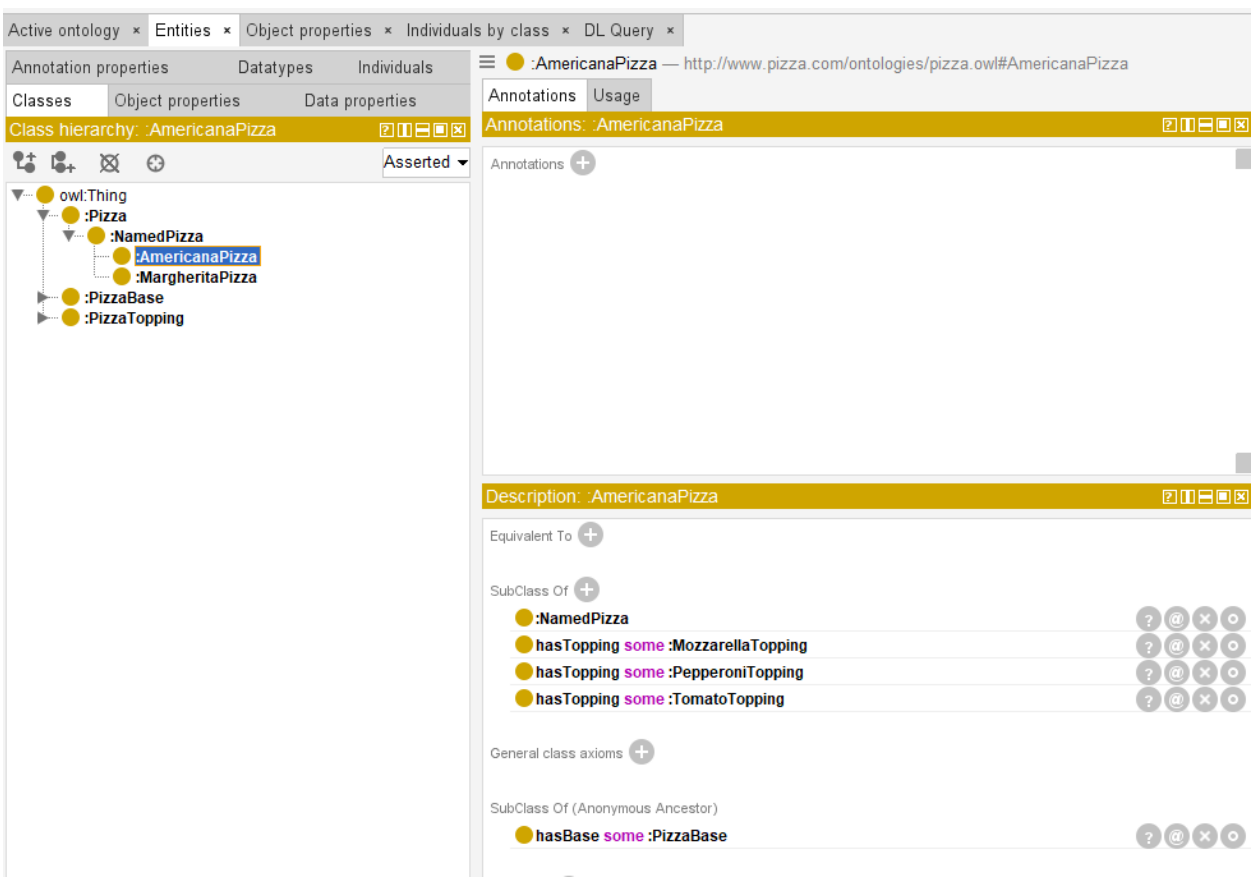
# Exercise 19


Active ontology
Entities
Object properties
Individuals by class
DL Query
Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Class hierarchy: :MargheritaPizza
Asserted
owl:Thing
:Pizza
:NamedPizza
:MargheritaPizza
:PizzaBase
:PizzaTopping
:CheeseTopping
:MozzarellaTopping
:ParmesanTopping
:MeatTopping
:SeafoodTopping
:VegetableTopping
:MargheritaPizza — http://www.pizza.com/ontolo
Annotations
Usage
Annotations: :MargheritaPizza
Annotations
rdfs:comment
A pizza that only has Mozzarella and Tomato toppings.
Description: :MargheritaPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping some :MozzarellaTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

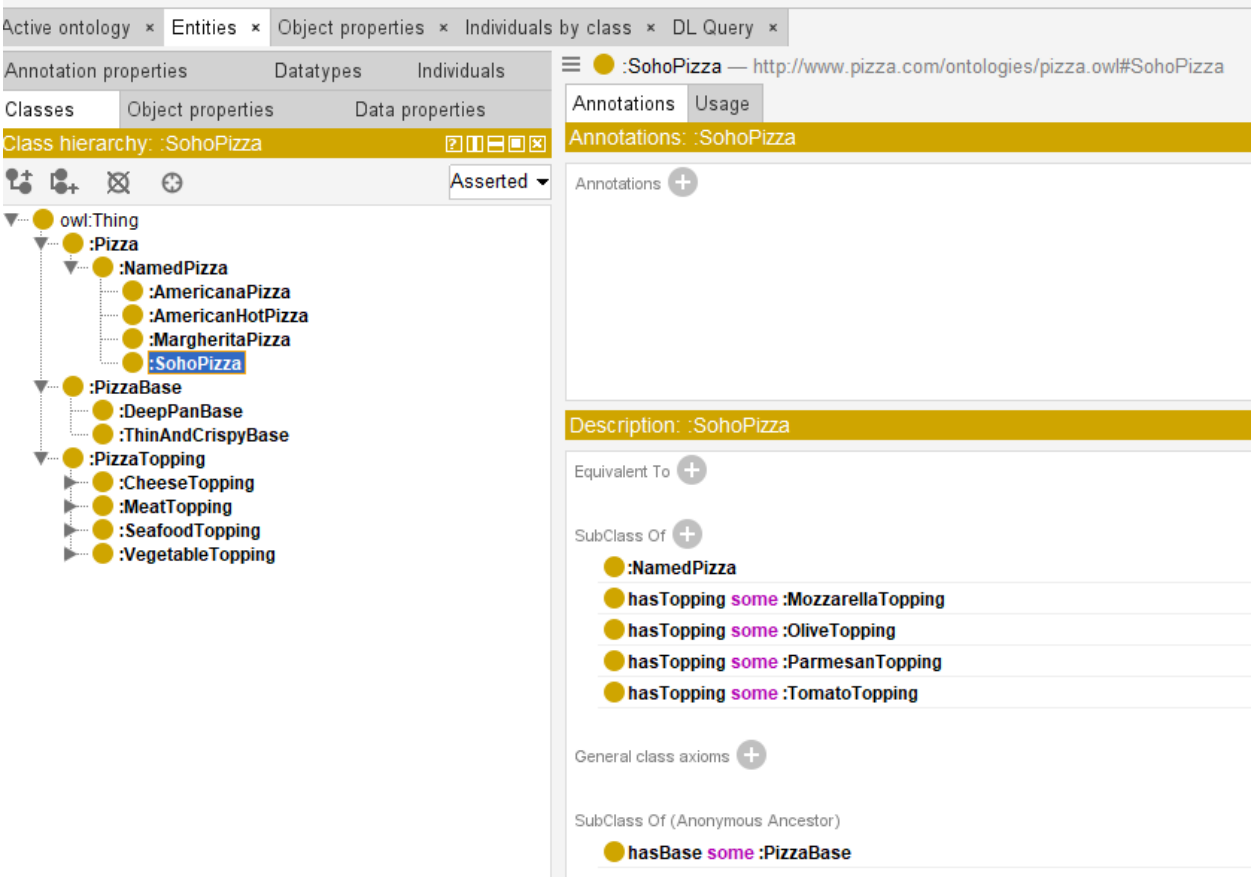
# Exercise 20


:Pizza :NamedPizza :MargheritaPizza
Active ontology
Entities
Object properties
Individuals by class
DL Query
Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Class hierarchy: :MargheritaPizza
Asserted
owl:Thing
:Pizza
:NamedPizza
:MargheritaPizza
:PizzaBase
:DeepPanBase
:ThinAndCrispyBase
:PizzaTopping
:CheeseTopping
:MeatTopping
:SeafoodTopping
:VegetableTopping
:MargheritaPizza — http://www.pizza.com/ontologies/pizza.owl#MargheritaPizza
Annotations
Usage
Annotations: :MargheritaPizza
Annotations
rdfs:comment
A pizza that only has Mozzarella and Tomato toppings.
Description: :MargheritaPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping some :MozzarellaTopping
hasTopping some :TomatoTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase
Instances

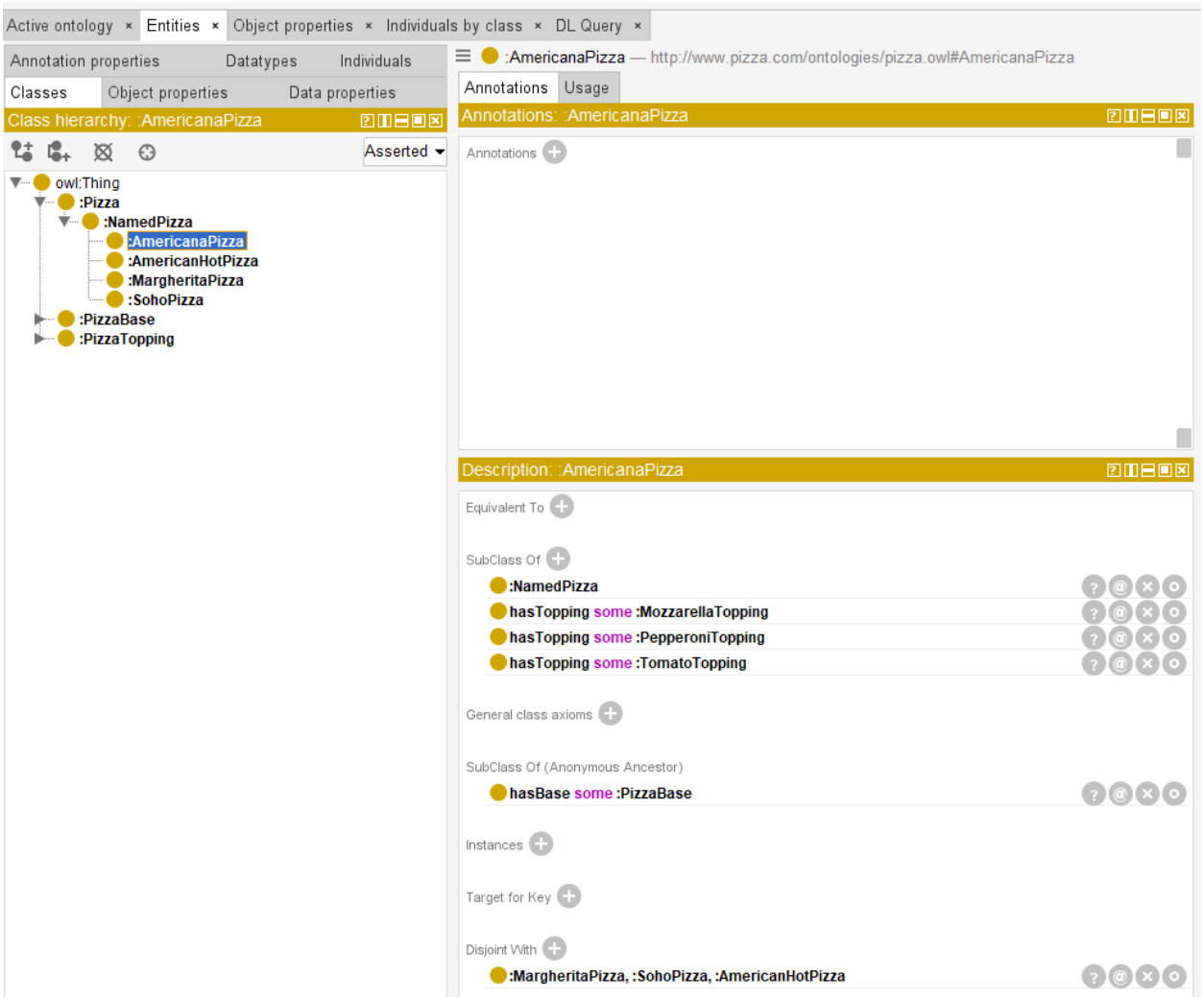
# Exercise 21


Active ontology
Entities
Object properties
Individuals by class
DL Query
Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Class hierarchy: :AmericanaPizza
Asserted
owl:Thing
:Pizza
:NamedPizza
:AmericanaPizza
:MargheritaPizza
:PizzaBase
:PizzaTopping
:AmericanaPizza — http://www.pizza.com/ontologies/pizza.owl#AmericanaPizza
Annotations
Usage
Annotations: :AmericanaPizza
Annotations
Description: :AmericanaPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping some :MozzarellaTopping
hasTopping some :PepperoniTopping
hasTopping some :TomatoTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

# Exercise 22




# Exercise 23


Active ontology
Entities
Object properties
Individuals by class
DL Query
Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Class hierarchy: :AmericanaPizza
Asserted
owl:Thing
:Pizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:PizzaBase
:PizzaTopping
:AmericanaPizza — http://www.pizza.com/ontologies/pizza.owl#AmericanaPizza
Annotations
Usage
Annotations: :AmericanaPizza
Annotations
Description: :AmericanaPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping some :MozzarellaTopping
hasTopping some :PepperoniTopping
hasTopping some :TomatoTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase
Instances
Target for Key
Disjoint With
:MargheritaPizza, :SohoPizza, :AmericanHotPizza

Class hierarchy: :AmericanHotPizza

Asserted

- owl:Thing
  - :Pizza
    - :NamedPizza
      - :AmericanaPizza
      - :AmericanHotPizza
      - :MargheritaPizza
      - :SohoPizza
  - :PizzaBase
  - :PizzaTopping

Annotations: :AmericanHotPizza

Annotations

Description: :AmericanHotPizza

Equivalent To

SubClass Of

- :NamedPizza
- hasTopping some :JalapenoPepperTopping
- hasTopping some :MozzarellaTopping
- hasTopping some :PepperoniTopping
- hasTopping some :TomatoTopping

General class axioms

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase

- owl:Thing
  - :Pizza
    - :NamedPizza
      - :AmericanaPizza
      - :AmericanHotPizza
      - :MargheritaPizza
      - :SohoPizza
  - :PizzaBase
  - :PizzaTopping

**rdfs:comment**

A pizza that only has Mozzarella and Tomato toppings.

**Description: :MargheritaPizza**

Equivalent To

SubClass Of

- :NamedPizza
- hasTopping some :MozzarellaTopping
- hasTopping some :TomatoTopping

General class axioms

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase

Instances

Target for Key

Disjoint With

- :AmericanaPizza, :SohoPizza, :AmericanHotPizza

Classes | Object properties | Data properties

**Class hierarchy: :SohoPizza**

Asserted

- owl:Thing
  - :Pizza
    - :NamedPizza
      - :AmericanaPizza
      - :AmericanHotPizza
      - :MargheritaPizza
      - :SohoPizza
  - :PizzaBase
  - :PizzaTopping

Annotations | Usage

**Annotations: :SohoPizza**

Annotations

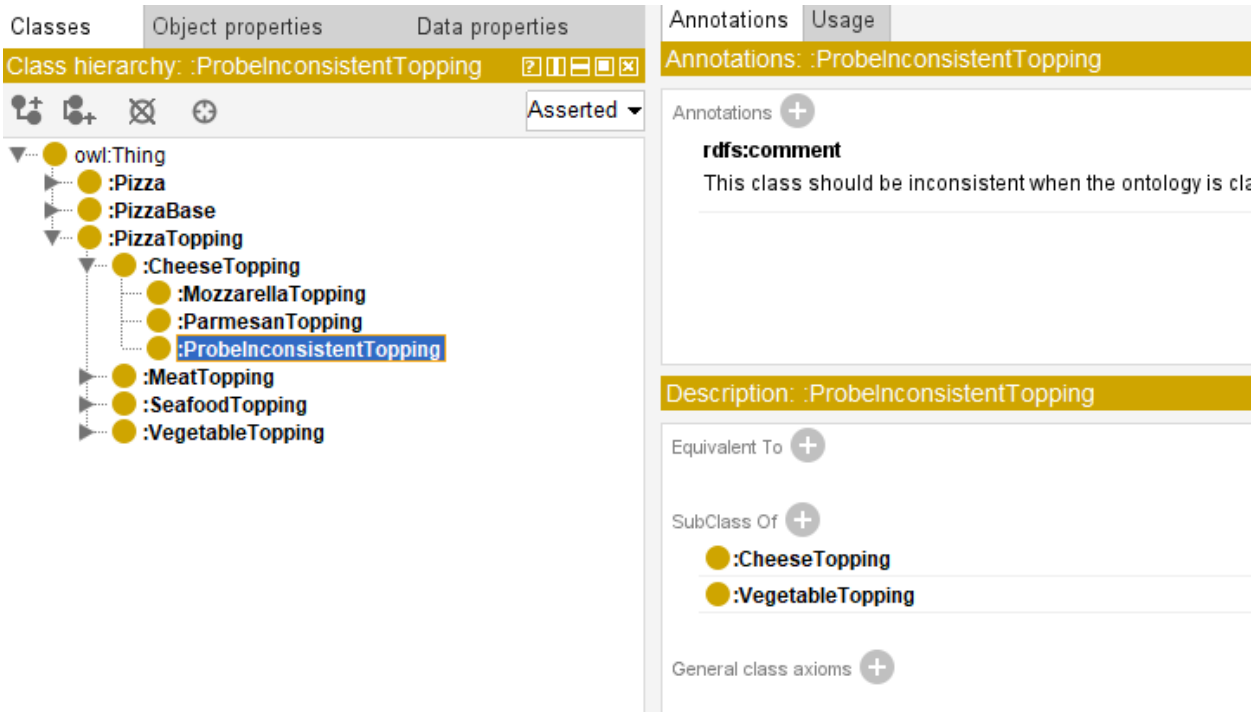
**Description: :SohoPizza**

Equivalent To

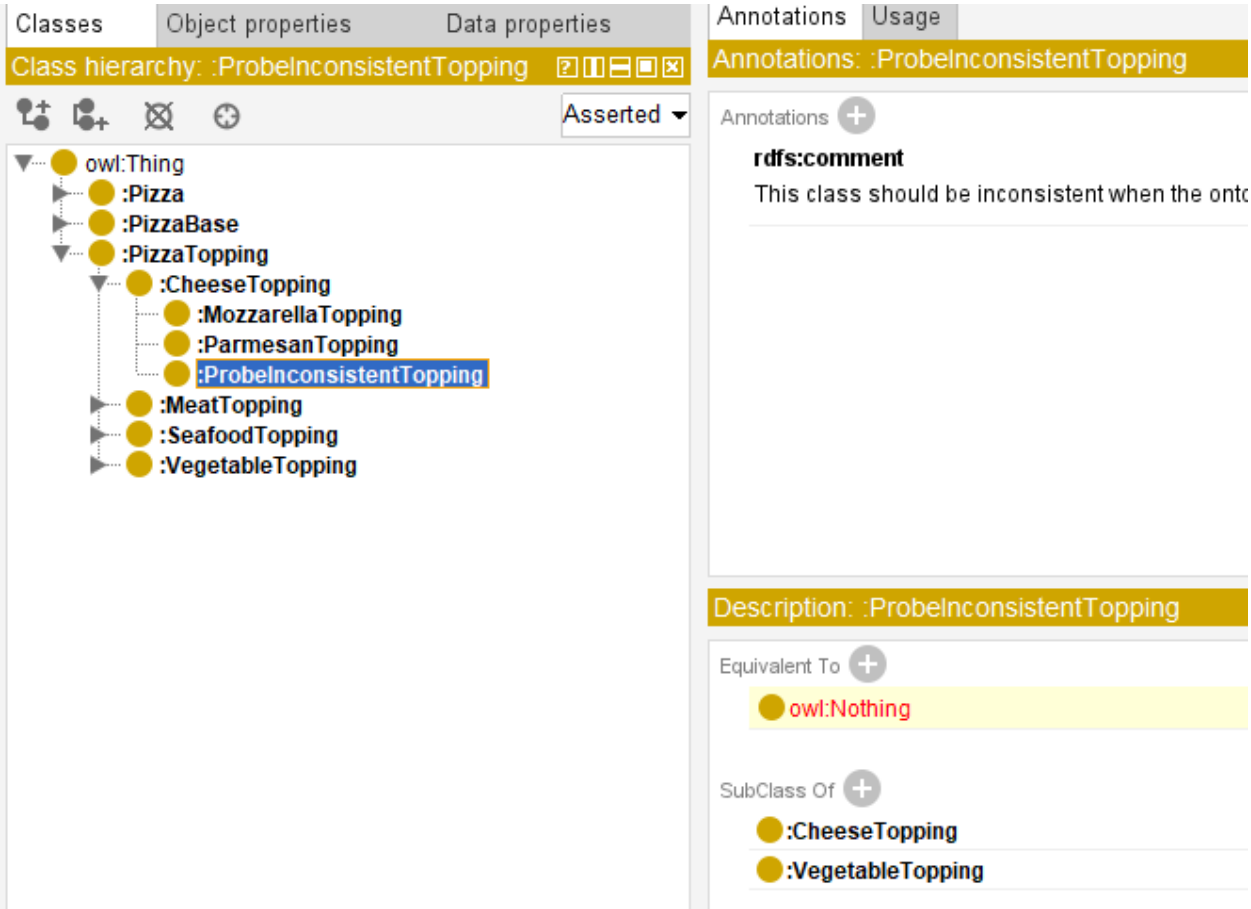
SubClass Of

- :NamedPizza
- hasTopping some :MozzarellaTopping
- hasTopping some :OliveTopping
- hasTopping some :ParmesanTopping
- hasTopping some :TomatoTopping

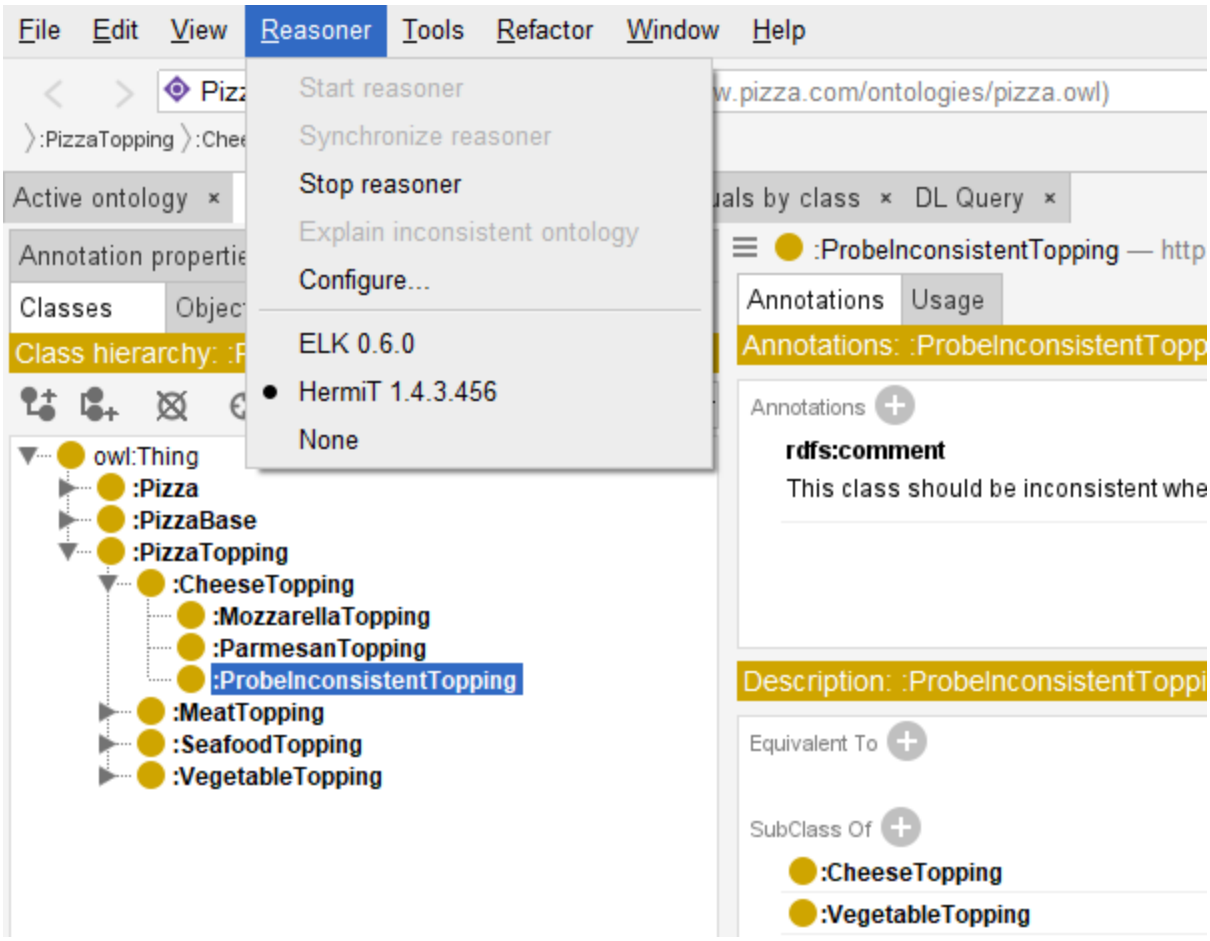
# Exercise 24



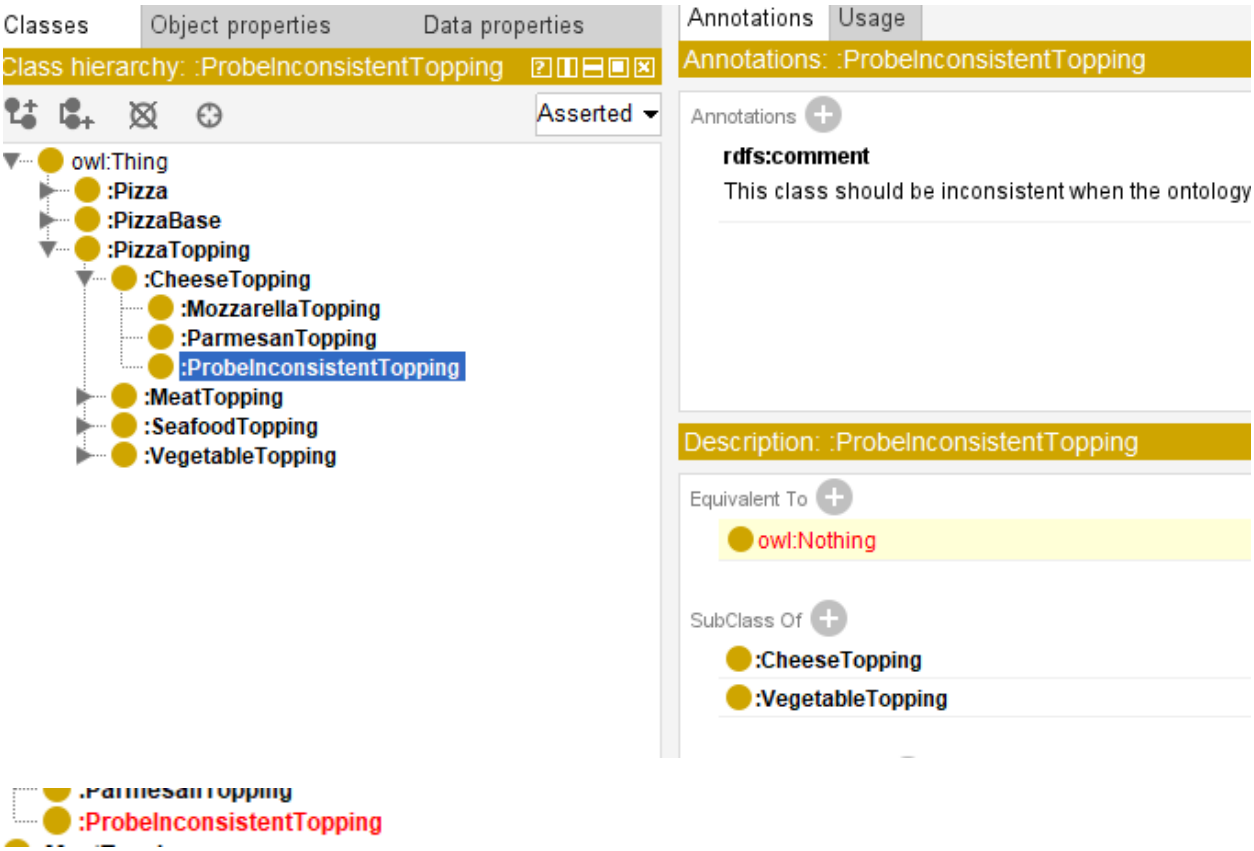

# Exercise 25


Classes
Object properties
Data properties
Class hierarchy: :ProbeInconsistentTopping
Asserted
owl:Thing
:Pizza
:PizzaBase
:PizzaTopping
:CheeseTopping
:MozzarellaTopping
:ParmesanTopping
:ProbeInconsistentTopping
:MeatTopping
:SeafoodTopping
:VegetableTopping
Annotations
Usage
Annotations: :ProbeInconsistentTopping
Annotations
rdfs:comment
This class should be inconsistent when the onto
Description: :ProbeInconsistentTopping
Equivalent To
owl:Nothing
SubClass Of
:CheeseTopping
:VegetableTopping

# Exercise 26

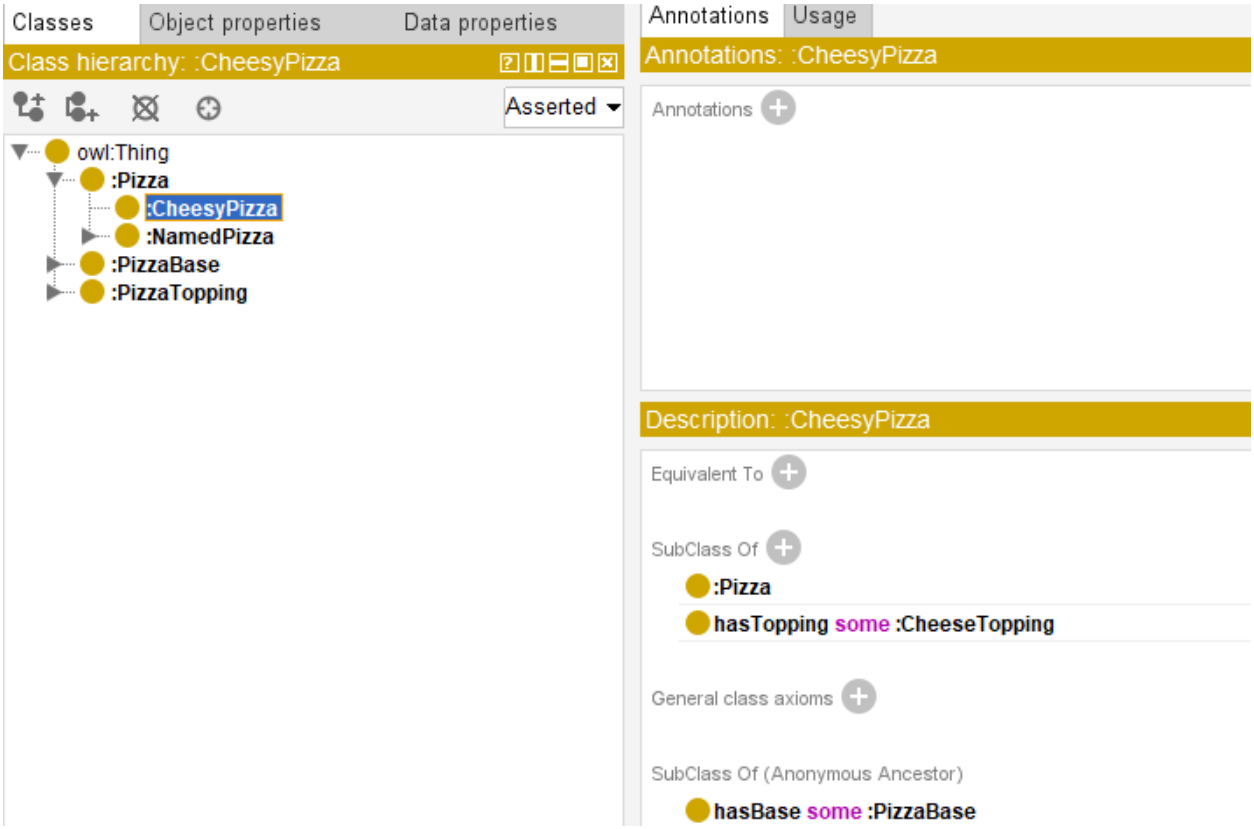

File
Edit
View
Reasoner
Tools
Refactor
Window
Help
Start reasoner
Synchronize reasoner
Stop reasoner
Explain inconsistent ontology
Configure...
ELK 0.6.0
HermiT 1.4.3.456
None
Active ontology
Classes
Class hierarchy:
owl:Thing
:Pizza
:PizzaBase
:PizzaTopping
:CheeseTopping
:MozzarellaTopping
:ParmesanTopping
:ProbeInconsistentTopping
:MeatTopping
:SeafoodTopping
:VegetableTopping
DL Query
:ProbeInconsistentTopping
Annotations
Usage
Annotations: :ProbeInconsistentTopp
Annotations
rdfs:comment
This class should be inconsistent whe
Description: :ProbeInconsistentToppi
Equivalent To
SubClass Of
:CheeseTopping
:VegetableTopping

# Exercise 27

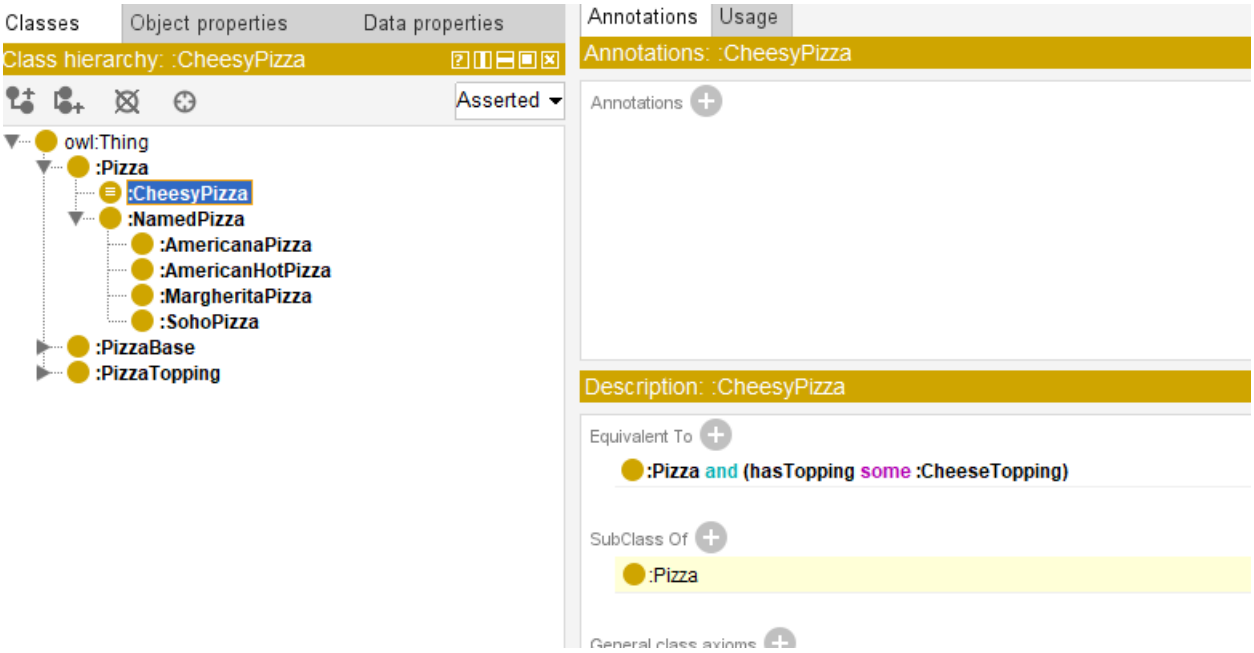

Classes
Object properties
Data properties
Class hierarchy: :ProbeInconsistentTopping
Asserted
owl:Thing
:Pizza
:PizzaBase
:PizzaTopping
:CheeseTopping
:MozzarellaTopping
:ParmesanTopping
:ProbeInconsistentTopping
:MeatTopping
:SeafoodTopping
:VegetableTopping
Annotations
Usage
Annotations: :ProbeInconsistentTopping
Annotations
rdfs:comment
This class should be inconsistent when the ontology
Description: :ProbeInconsistentTopping
Equivalent To
owl:Nothing
SubClass Of
:CheeseTopping
:VegetableTopping


:ProbeInconsistentTopping

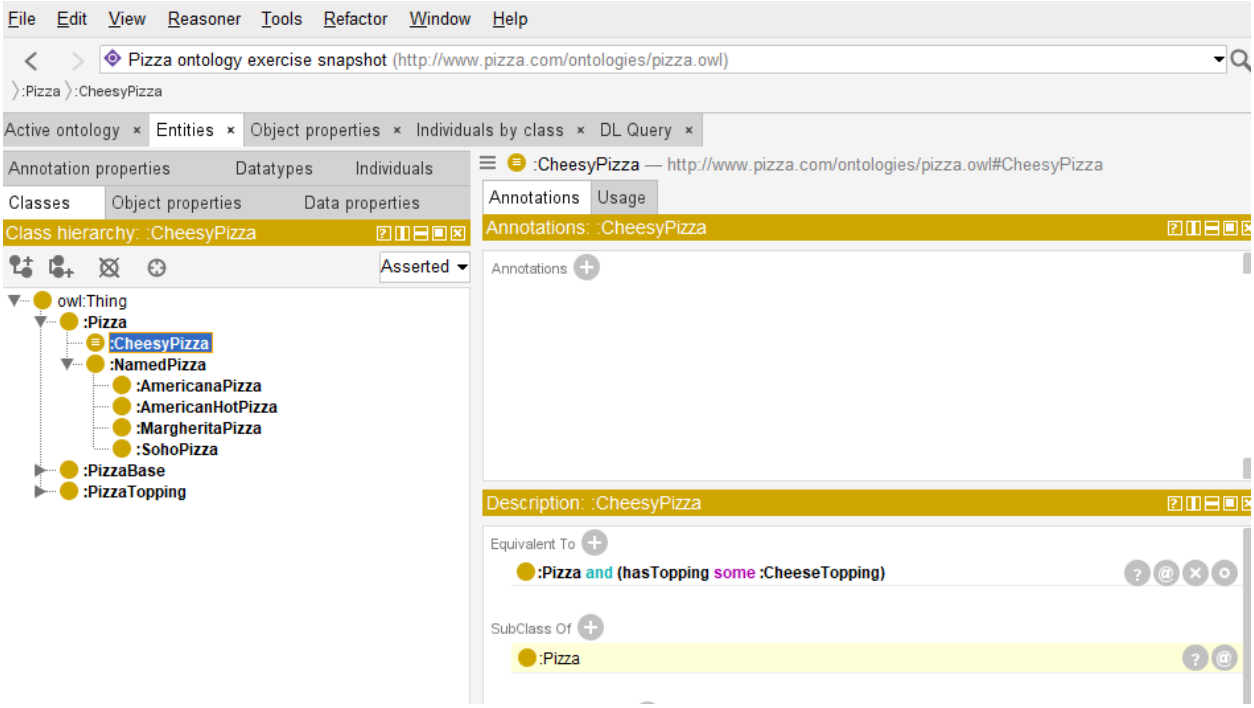
# Exercise 28



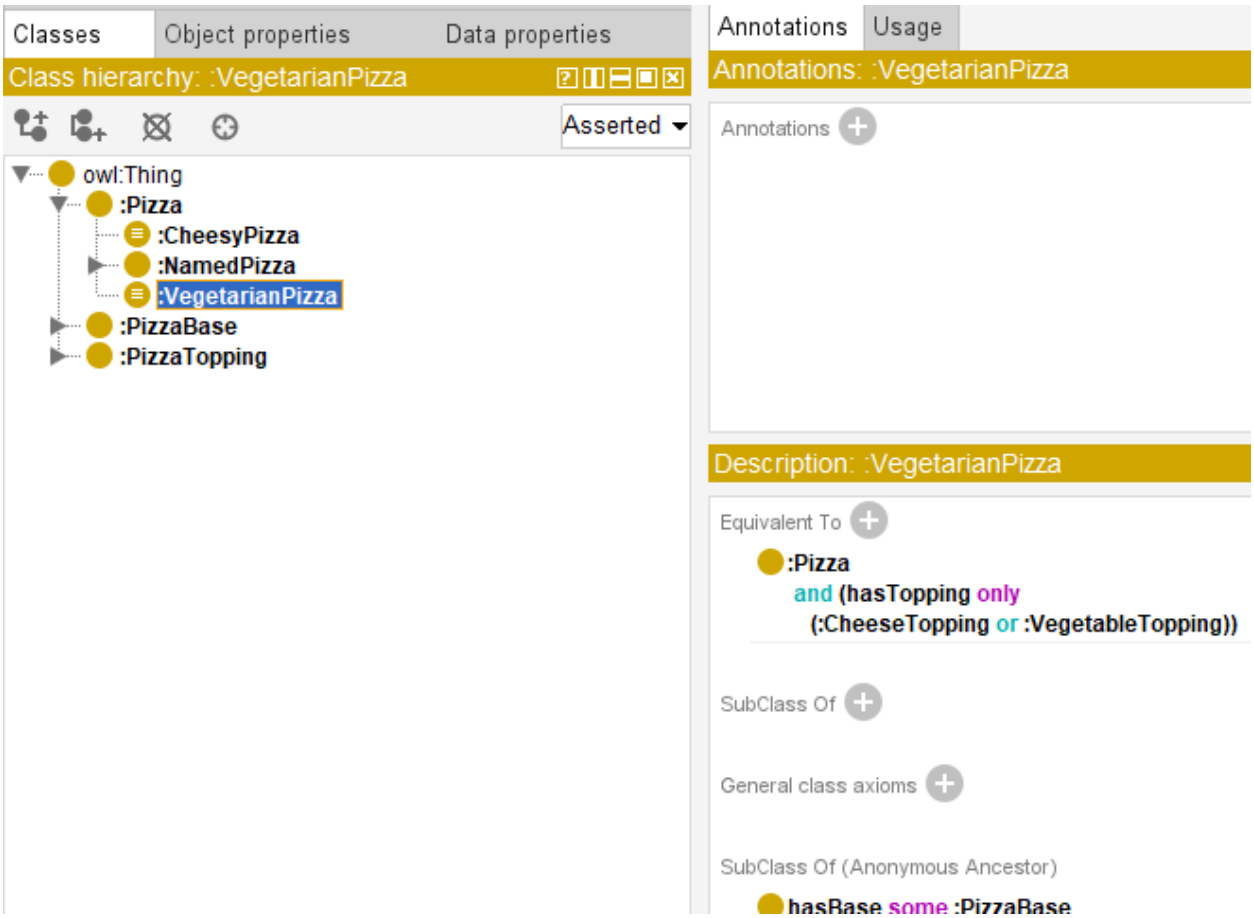

# Exercise 29



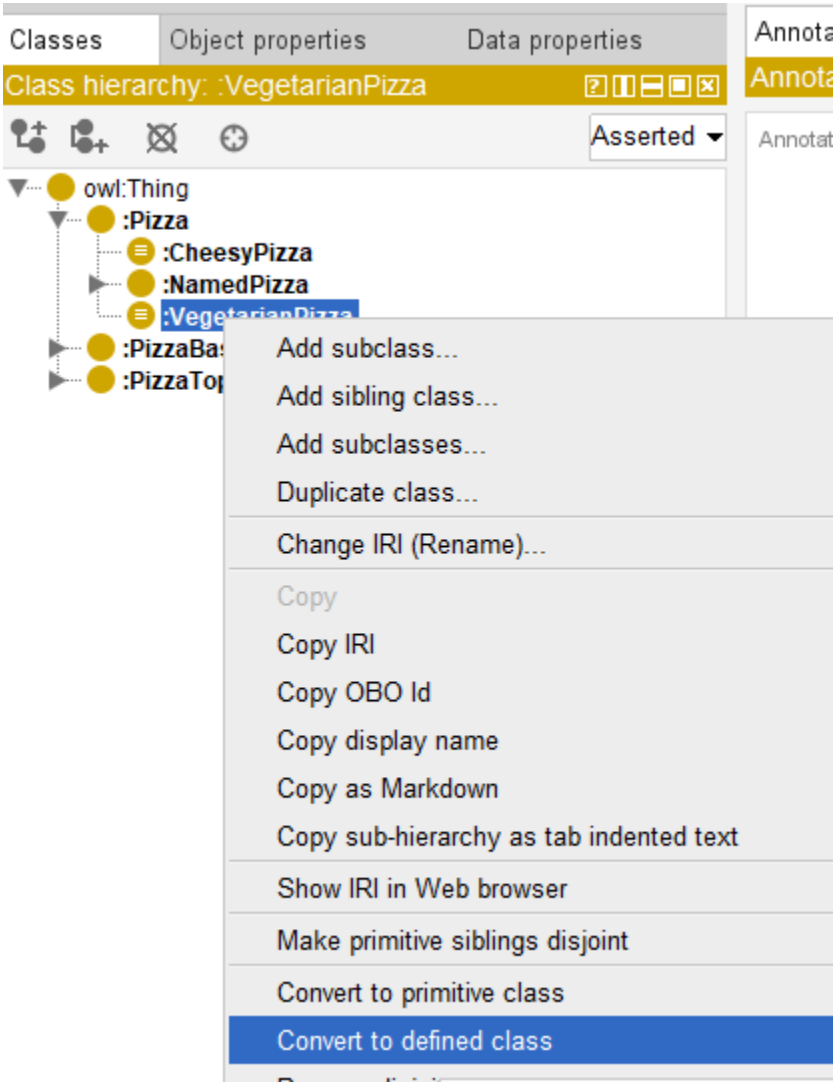

# Exercise 30


File Edit View Reasoner Tools Refactor Window Help
Pizza ontology exercise snapshot (http://www.pizza.com/ontologies/pizza.owl)
:Pizza :CheesyPizza
Active ontology Entities Object properties Individuals by class DL Query
Annotation properties Datatypes Individuals
Classes Object properties Data properties
Class hierarchy: :CheesyPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:PizzaBase
:PizzaTopping
:CheesyPizza — http://www.pizza.com/ontologies/pizza.owl#CheesyPizza
Annotations Usage
Annotations: :CheesyPizza
Annotations
Description: :CheesyPizza
Equivalent To
:Pizza and (hasTopping some :CheeseTopping)
SubClass Of
:Pizza

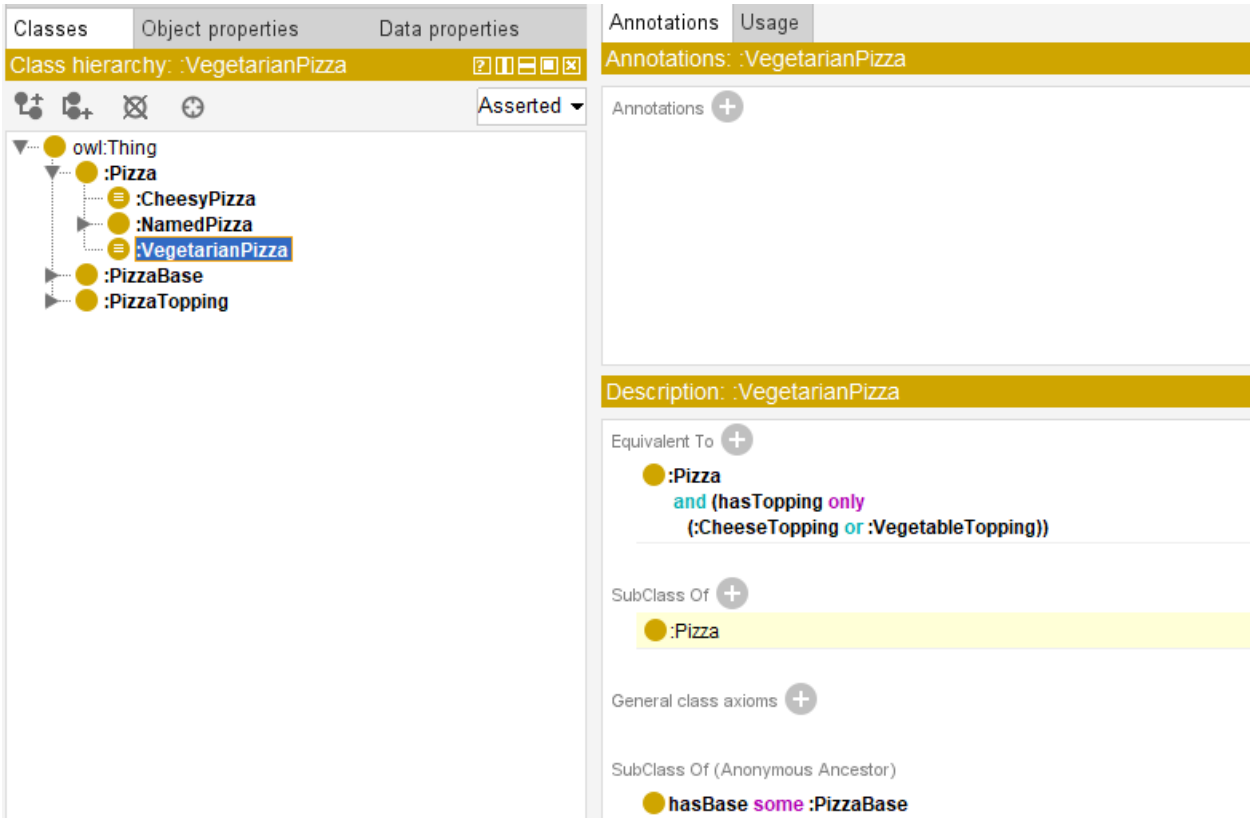
# Exercise 31


Classes Object properties Data properties
Class hierarchy: :VegetarianPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
Annotations Usage
Annotations: :VegetarianPizza
Annotations
Description: :VegetarianPizza
Equivalent To
:Pizza
and (hasTopping only
(:CheeseTopping or :VegetableTopping))
SubClass Of
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

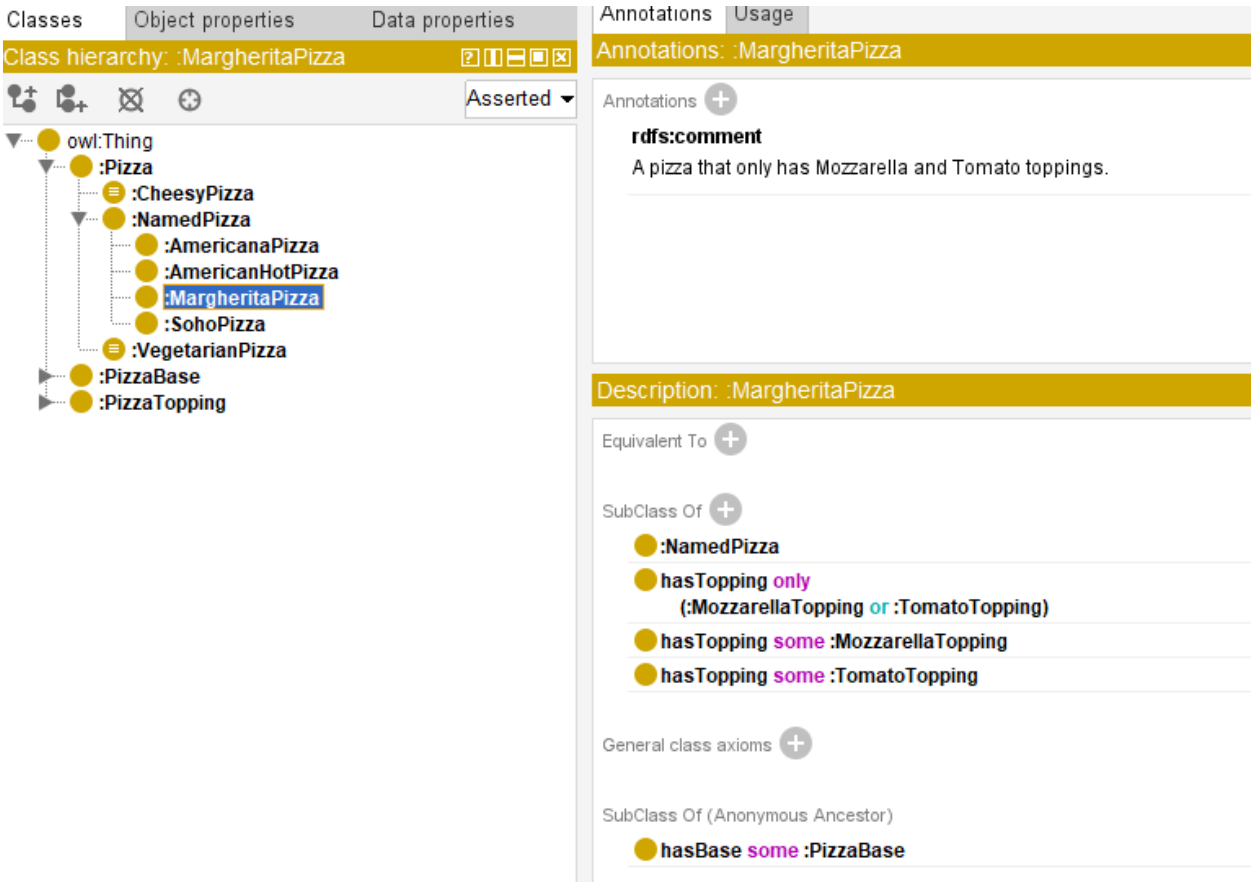
# Exercise 32



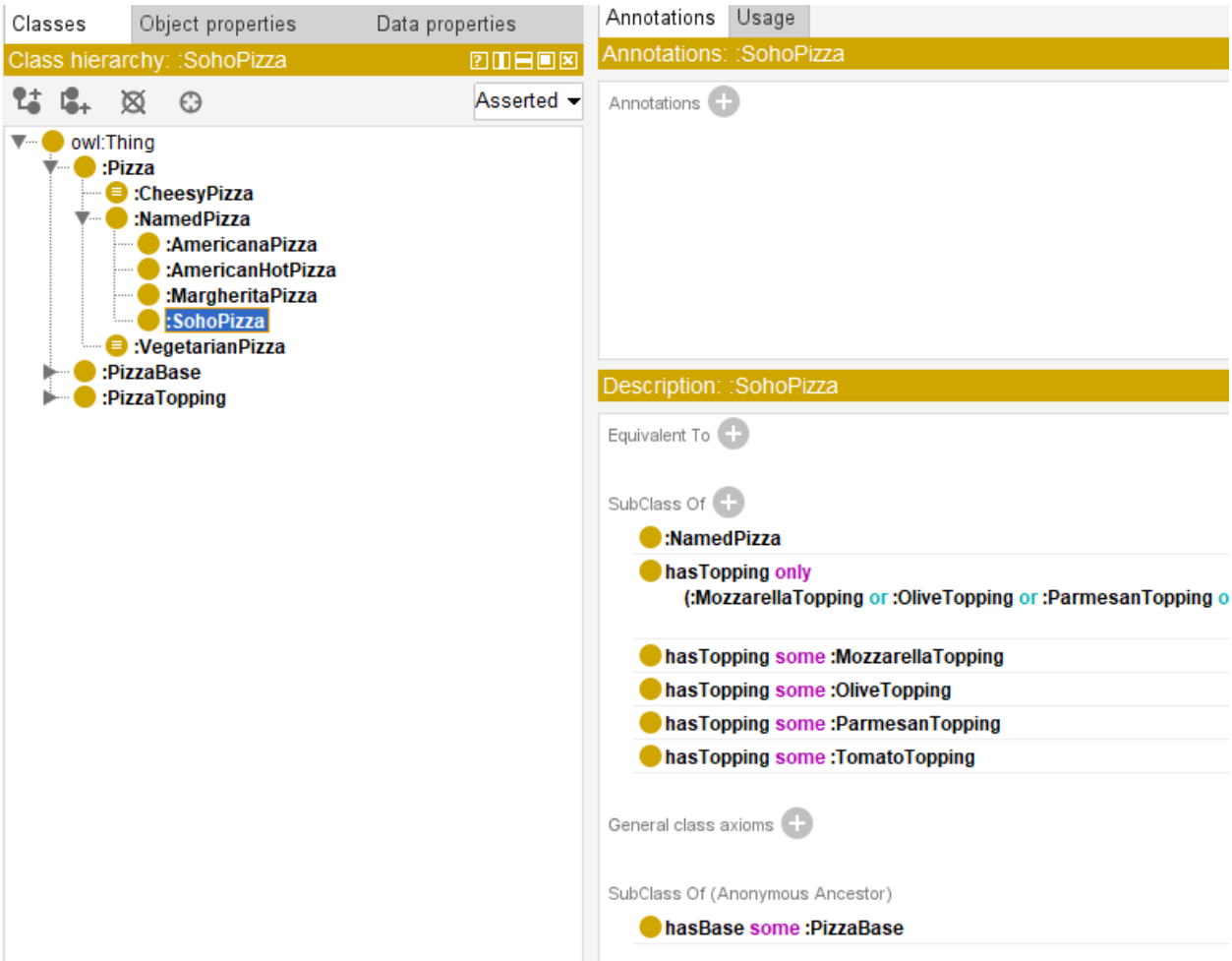

# Exercise 33


Classes
Object properties
Data properties
Class hierarchy: :VegetarianPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
Annotations
Usage
Annotations: :VegetarianPizza
Annotations
Description: :VegetarianPizza
Equivalent To
:Pizza
and (hasTopping only
(:CheeseTopping or :VegetableTopping))
SubClass Of
:Pizza
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

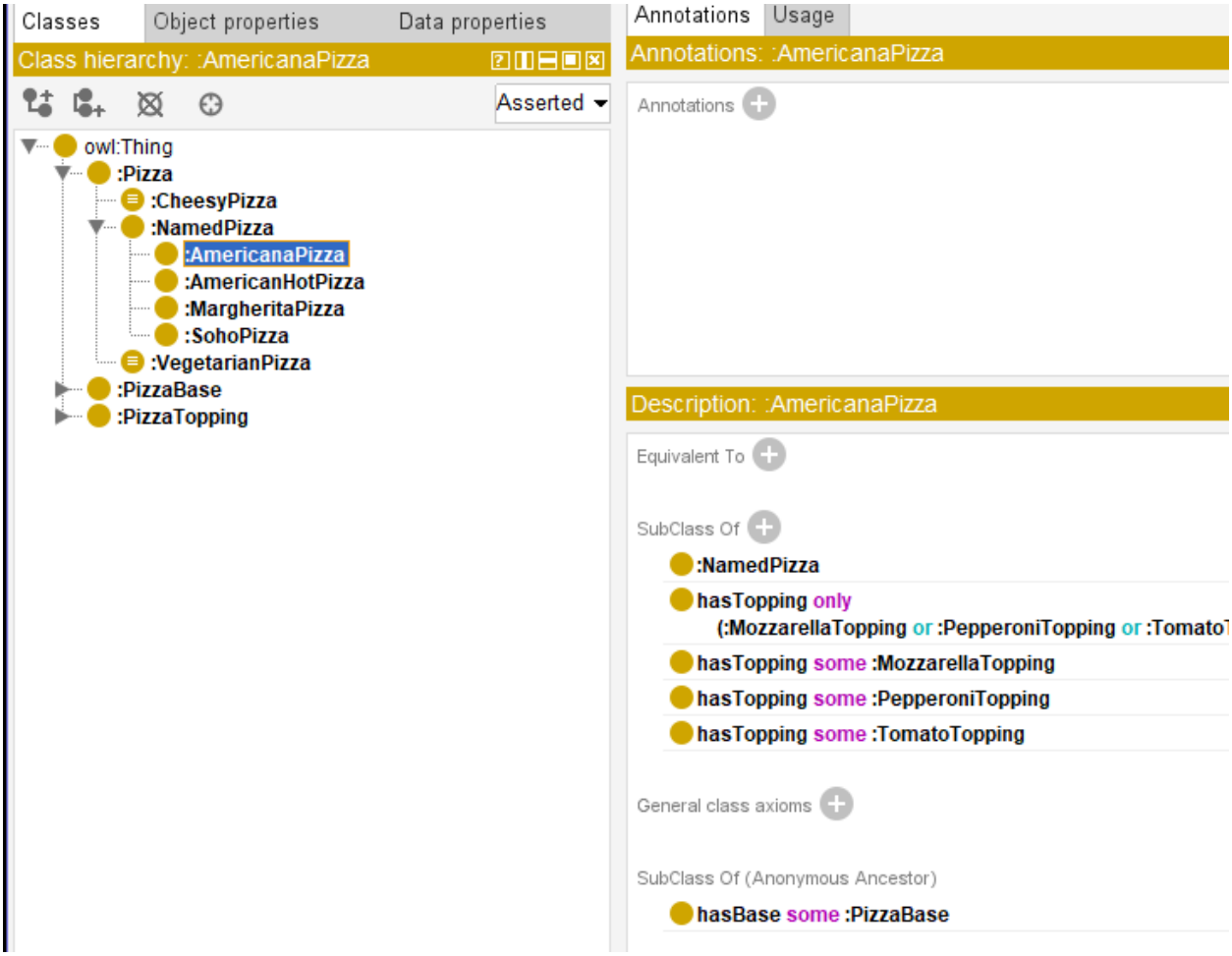
# Exercise 34


Classes
Object properties
Data properties
Class hierarchy: :MargheritaPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
Annotations
Usage
Annotations: :MargheritaPizza
Annotations
rdfs:comment
A pizza that only has Mozzarella and Tomato toppings.
Description: :MargheritaPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping only
(:MozzarellaTopping or :TomatoTopping)
hasTopping some :MozzarellaTopping
hasTopping some :TomatoTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

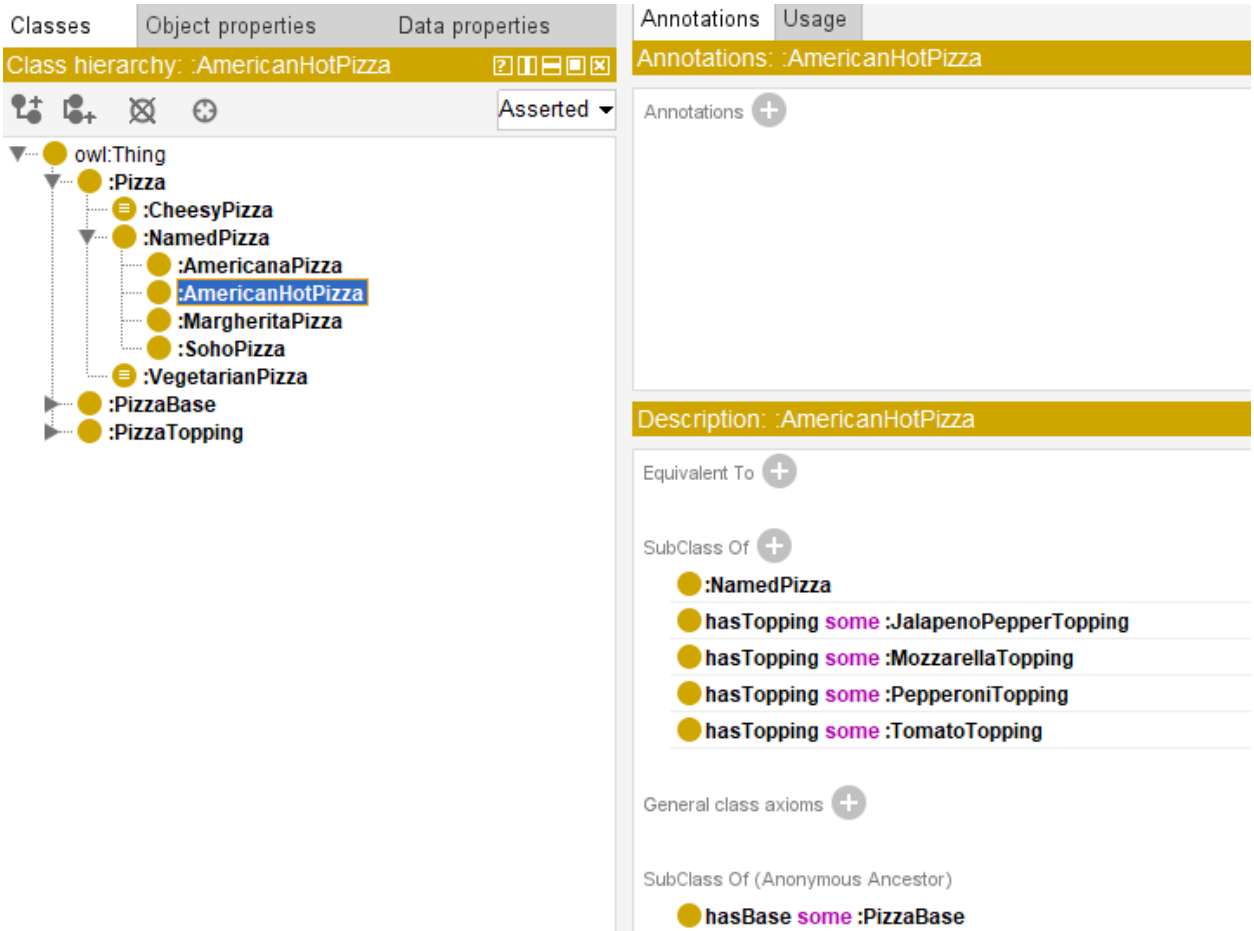
# Exercise 35


Classes
Object properties
Data properties
Class hierarchy: :SohoPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
Annotations
Usage
Annotations: :SohoPizza
Annotations
Description: :SohoPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping only
(:MozzarellaTopping or :OliveTopping or :ParmesanTopping o
hasTopping some :MozzarellaTopping
hasTopping some :OliveTopping
hasTopping some :ParmesanTopping
hasTopping some :TomatoTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

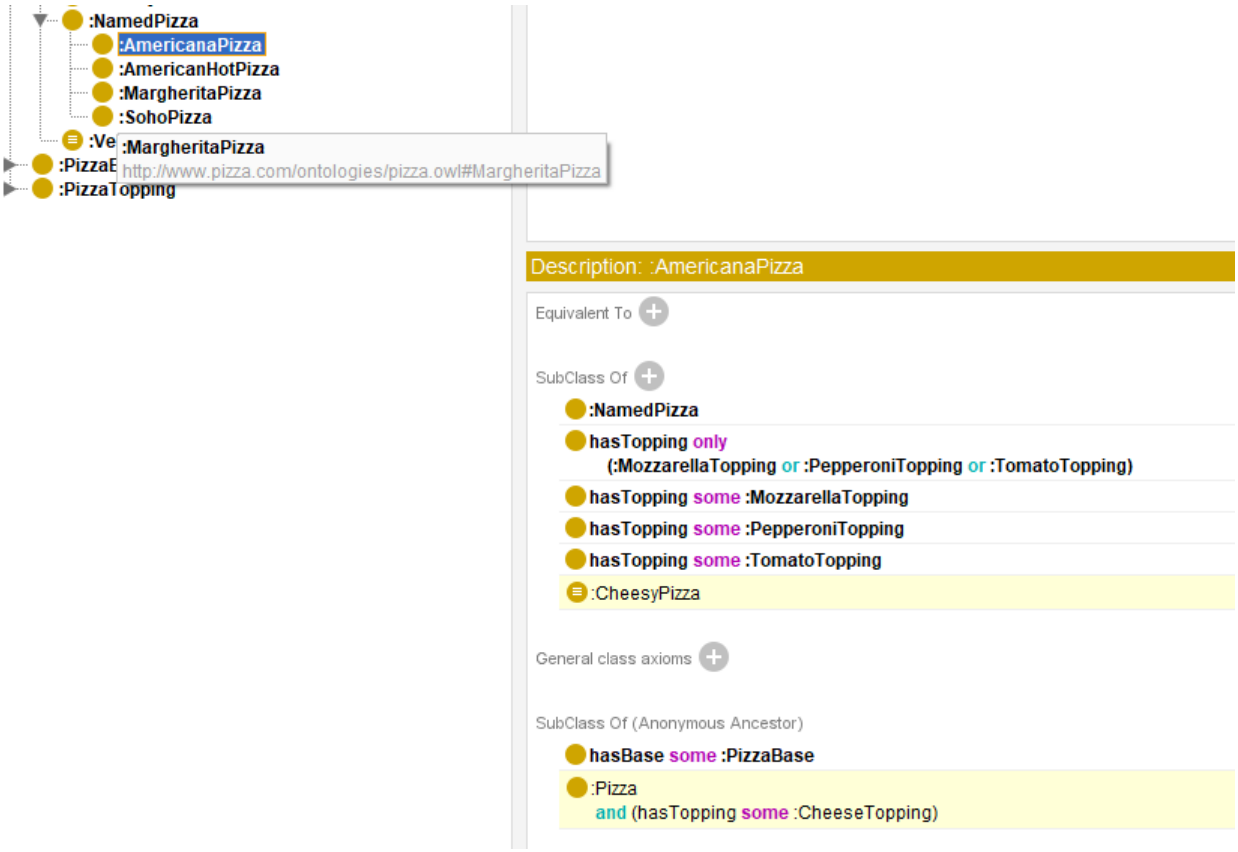
# Exercise 36


Classes
Object properties
Data properties
Class hierarchy: :AmericanaPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
Annotations
Usage
Annotations: :AmericanaPizza
Annotations
Description: :AmericanaPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping only
(:MozzarellaTopping or :PepperoniTopping or :Tomato
hasTopping some :MozzarellaTopping
hasTopping some :PepperoniTopping
hasTopping some :TomatoTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

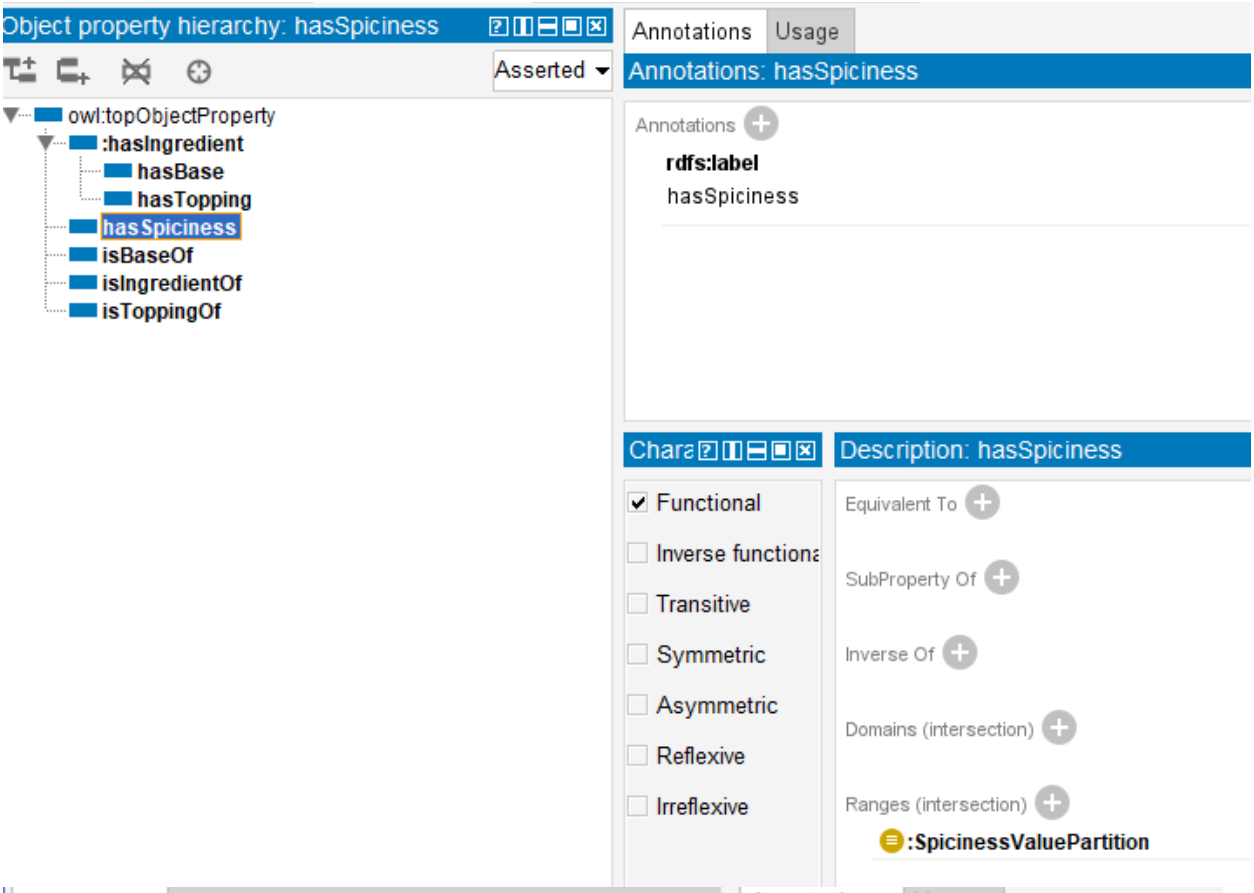
# Exercise 37

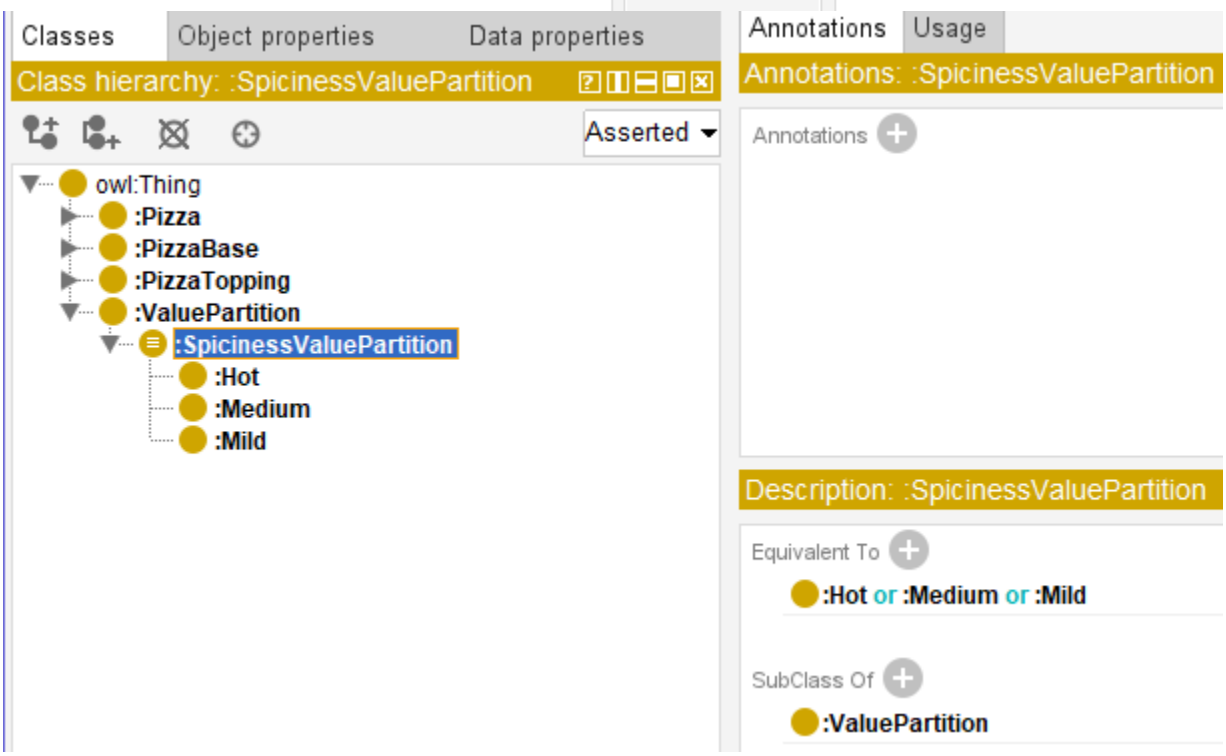

Classes
Object properties
Data properties
Class hierarchy: :AmericanHotPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
Annotations
Usage
Annotations: :AmericanHotPizza
Annotations
Description: :AmericanHotPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping some :JalapenoPepperTopping
hasTopping some :MozzarellaTopping
hasTopping some :PepperoniTopping
hasTopping some :TomatoTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

# Exercise 38

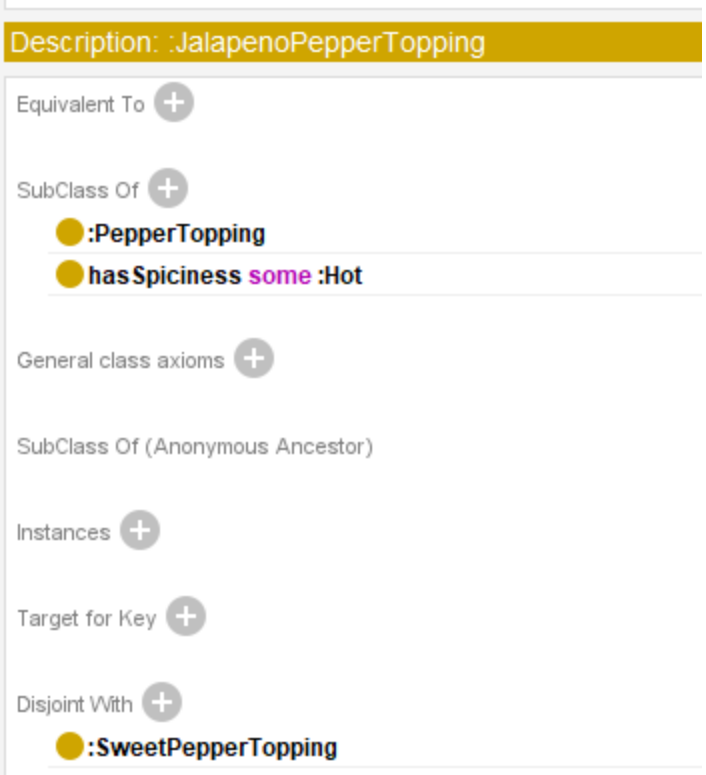

:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:MargheritaPizza
http://www.pizza.com/ontologies/pizza.owl#MargheritaPizza
:PizzaTopping
Description: :AmericanaPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping only
(:MozzarellaTopping or :PepperoniTopping or :TomatoTopping)
hasTopping some :MozzarellaTopping
hasTopping some :PepperoniTopping
hasTopping some :TomatoTopping
:CheesyPizza
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase
:Pizza
and (hasTopping some :CheeseTopping)

# Exercise 39

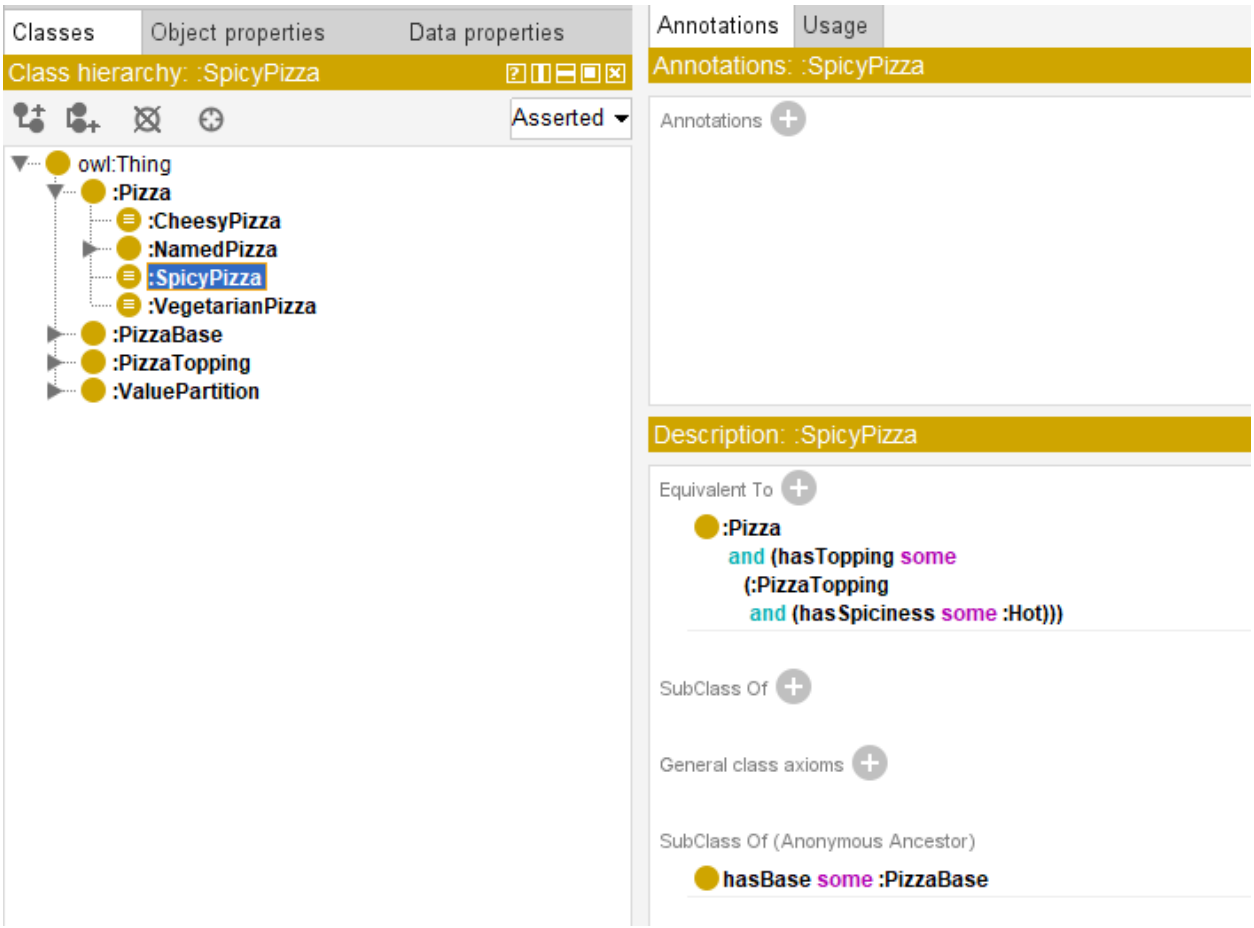

Object property hierarchy: hasSpiciness
Asserted
owl:topObjectProperty
:hasIngredient
hasBase
hasTopping
hasSpiciness
isBaseOf
isIngredientOf
isToppingOf
Annotations
Usage
Annotations: hasSpiciness
Annotations
rdfs:label
hasSpiciness
Chara
Description: hasSpiciness
Functional
Inverse functiona
Transitive
Symmetric
Asymmetric
Reflexive
Irreflexive
Equivalent To
SubProperty Of
Inverse Of
Domains (intersection)
Ranges (intersection)
:SpicinessValuePartition


Classes
Object properties
Data properties
Class hierarchy: :SpicinessValuePartition
Asserted
owl:Thing
:Pizza
:PizzaBase
:PizzaTopping
:ValuePartition
:SpicinessValuePartition
:Hot
:Medium
:Mild
Annotations
Usage
Annotations: :SpicinessValuePartition
Annotations
Description: :SpicinessValuePartition
Equivalent To
:Hot or :Medium or :Mild
SubClass Of
:ValuePartition

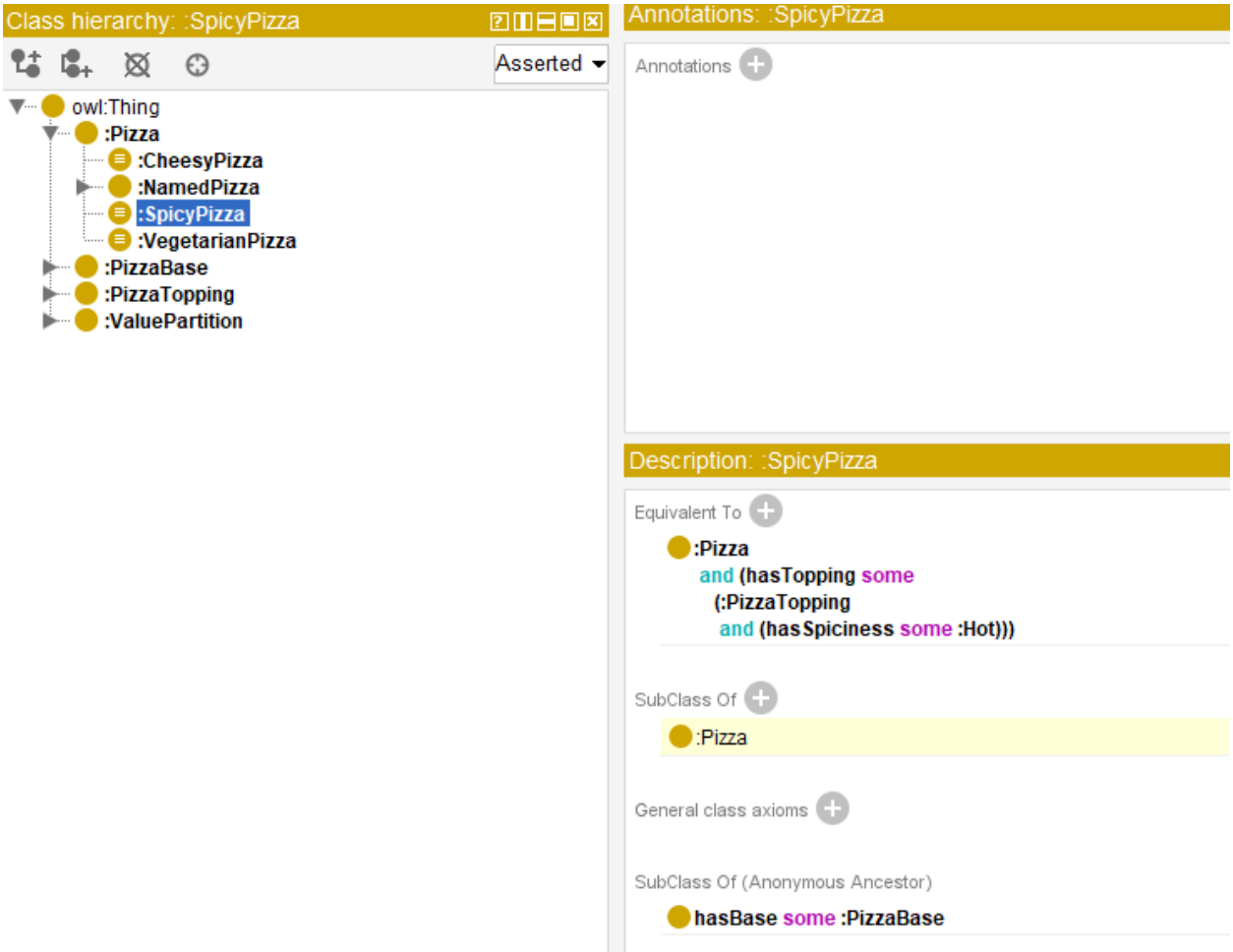
# Exercise 40


Description: :JalapenoPepperTopping
Equivalent To
SubClass Of
:PepperTopping
hasSpiciness some :Hot
General class axioms
SubClass Of (Anonymous Ancestor)
Instances
Target for Key
Disjoint With
:SweetPepperTopping

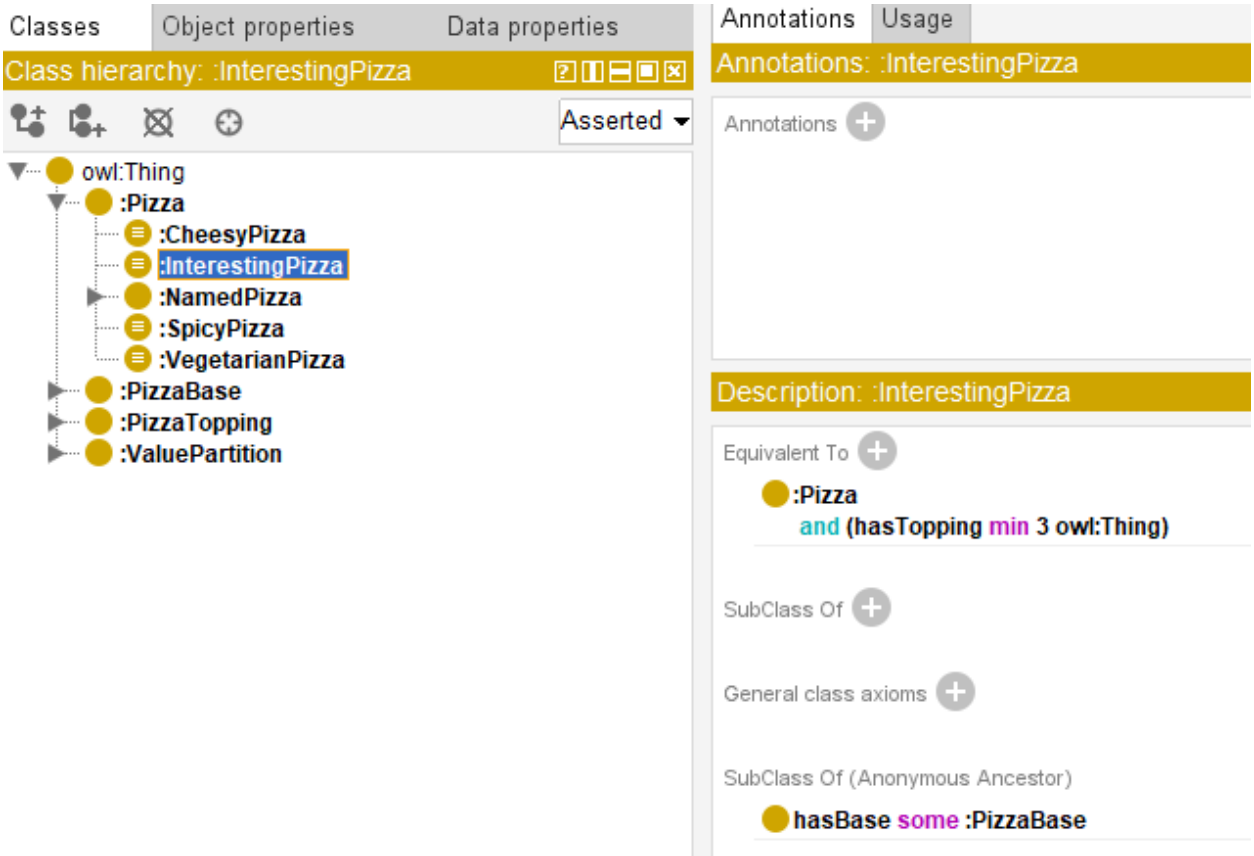
# Exercise 41


Classes
Object properties
Data properties
Class hierarchy: :SpicyPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :SpicyPizza
Annotations
Description: :SpicyPizza
Equivalent To
:Pizza
and (hasTopping some
(:PizzaTopping
and (hasSpiciness some :Hot)))
SubClass Of
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

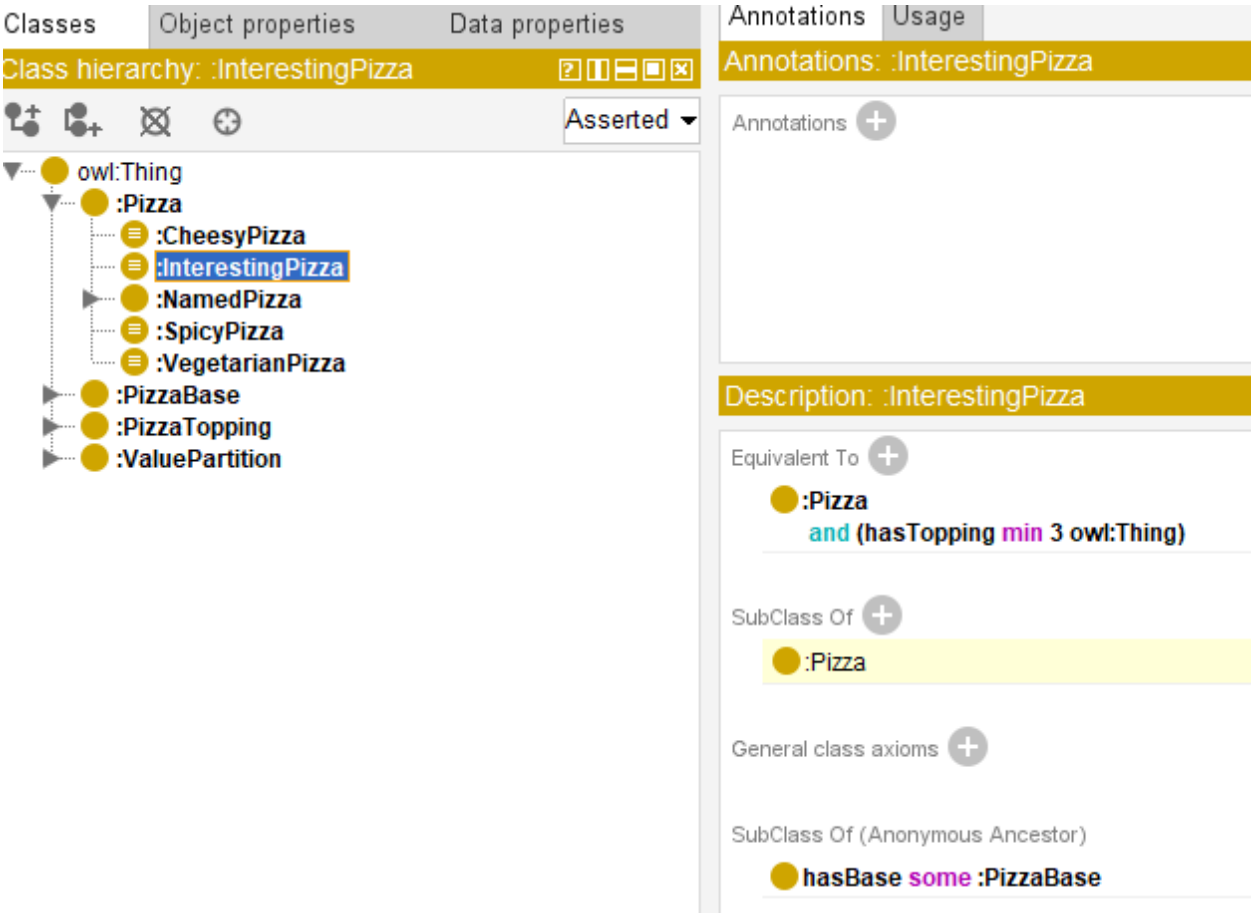
# Exercise 42


Class hierarchy: :SpicyPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:NamedPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations: :SpicyPizza
Annotations
Description: :SpicyPizza
Equivalent To
:Pizza
and (hasTopping some
(:PizzaTopping
and (hasSpiciness some :Hot)))
SubClass Of
:Pizza
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

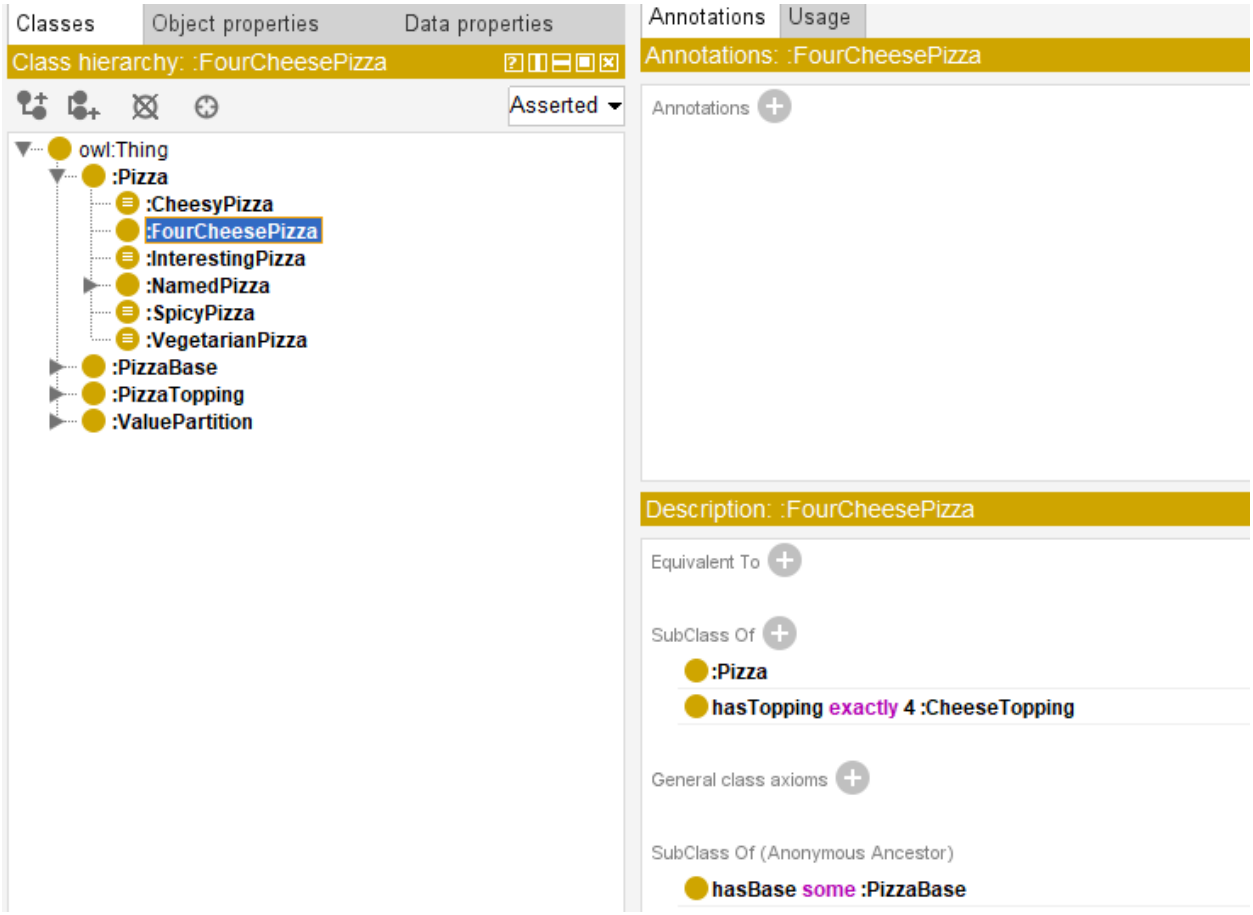
# Exercise 43


Classes
Object properties
Data properties
Class hierarchy: :Interesting Pizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:InterestingPizza
:NamedPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :InterestingPizza
Annotations
Description: :InterestingPizza
Equivalent To
:Pizza
and (hasTopping min 3 owl:Thing)
SubClass Of
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

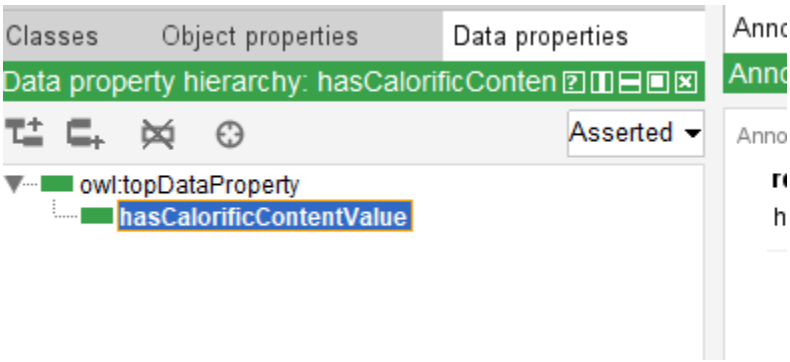
# Exercise 44


Classes
Object properties
Data properties
Class hierarchy: :InterestingPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:InterestingPizza
:NamedPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :InterestingPizza
Annotations
Description: :InterestingPizza
Equivalent To
:Pizza
and (hasTopping min 3 owl:Thing)
SubClass Of
:Pizza
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase

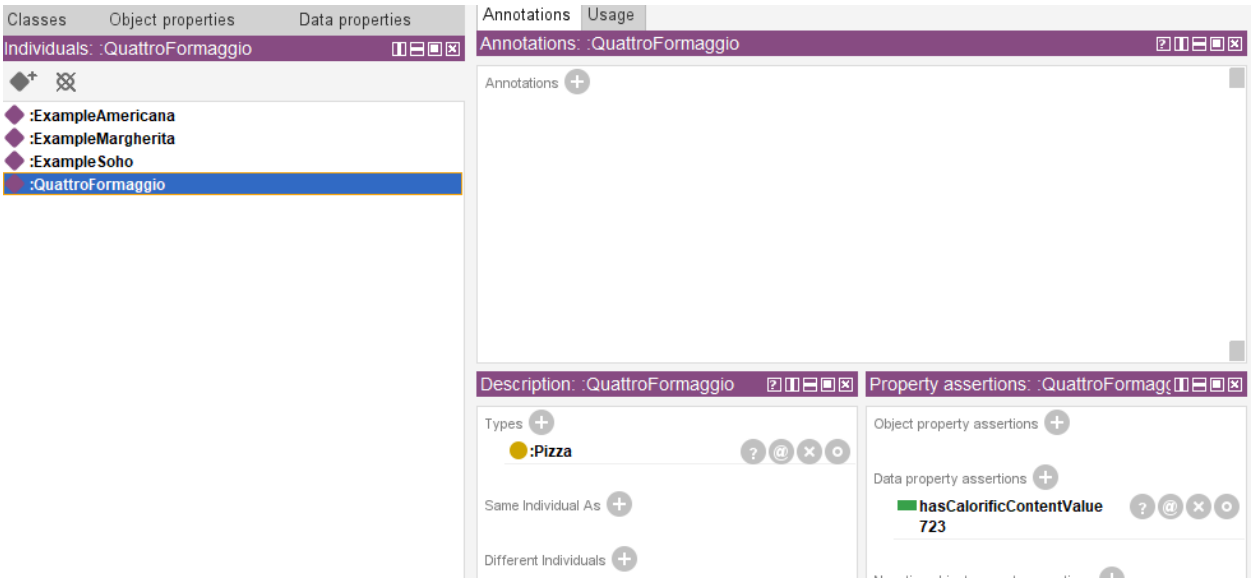
# Exercise 45



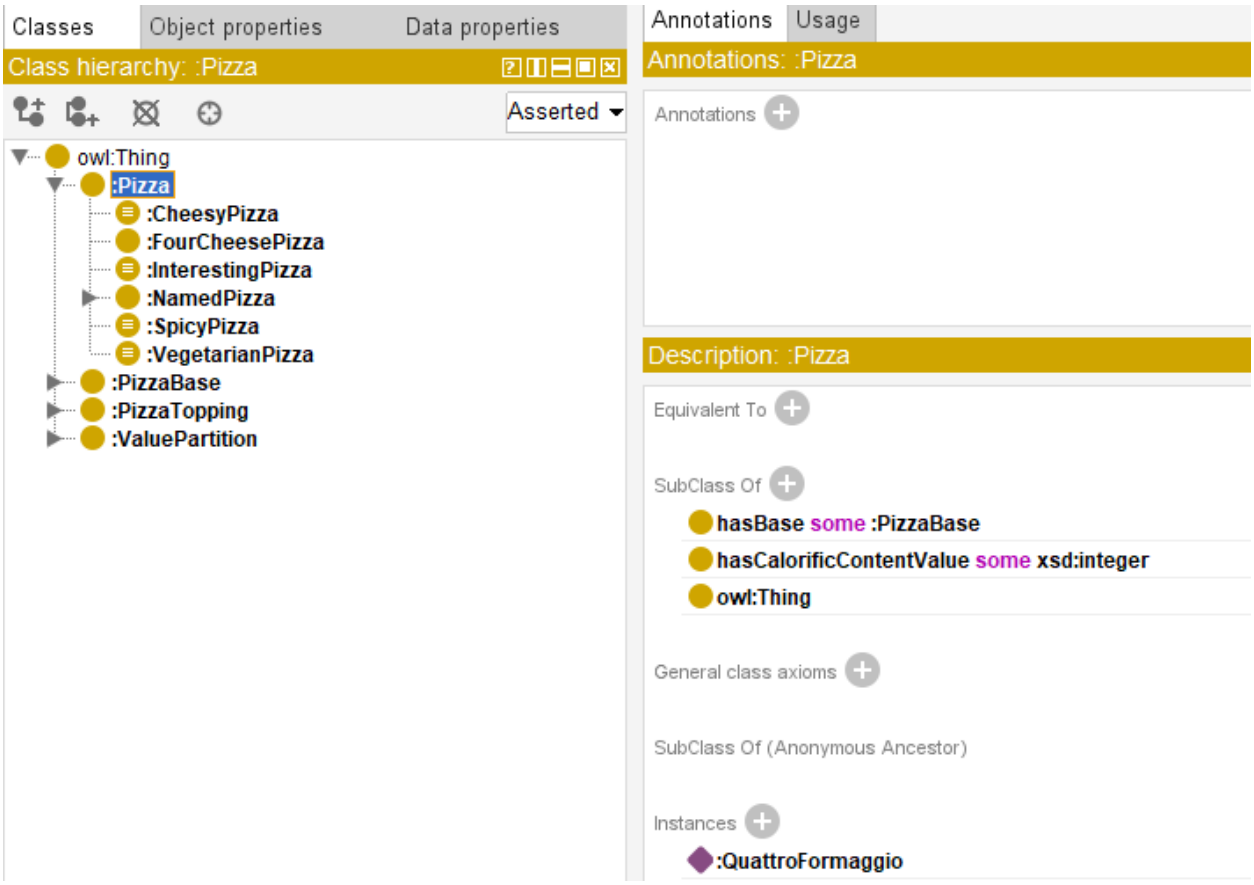

# Exercise 46



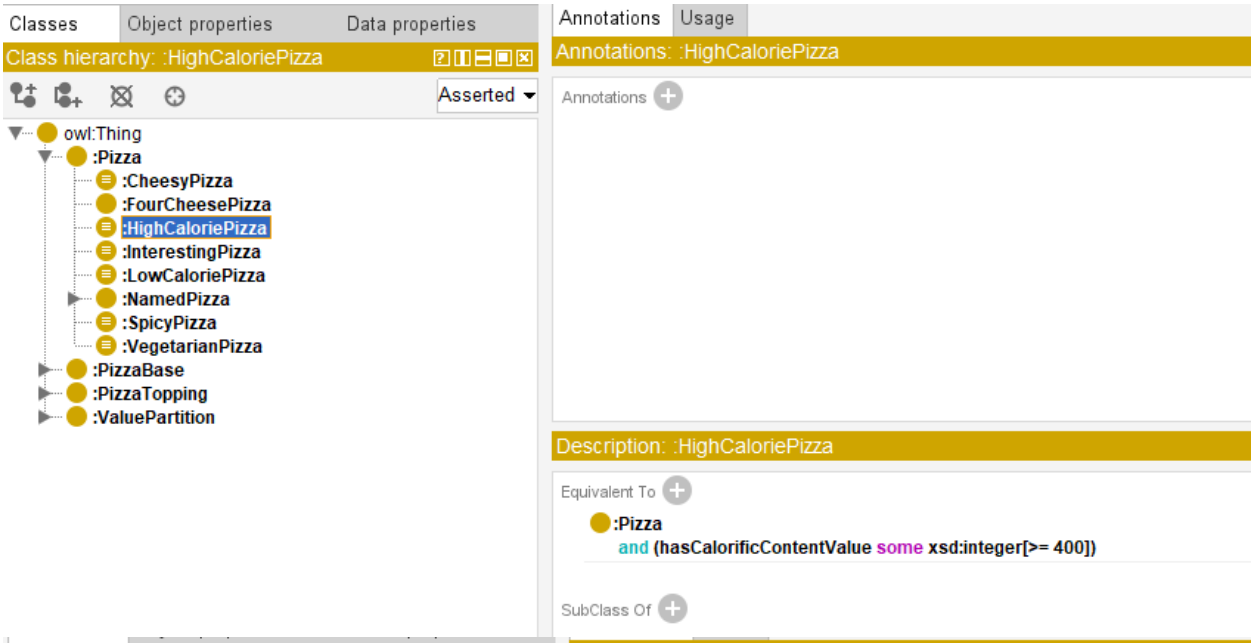

# Exercise 47

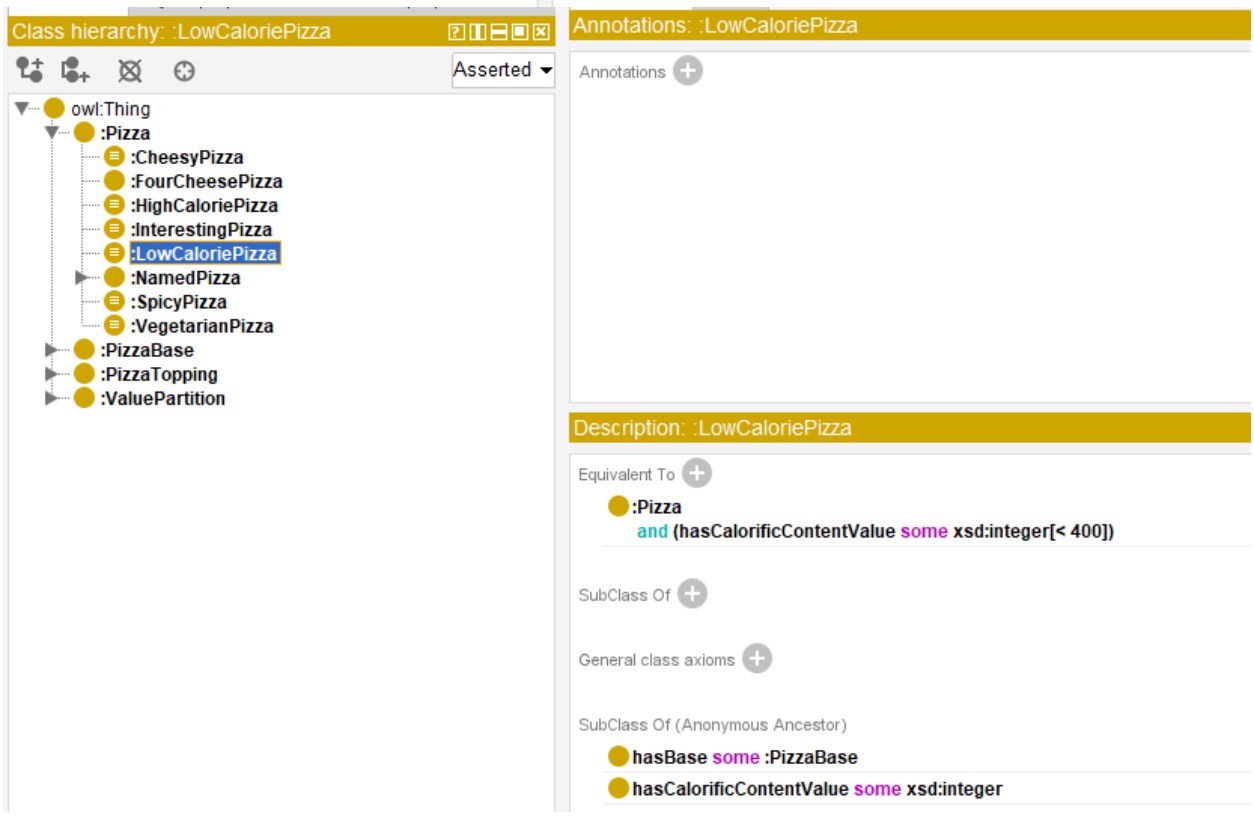



# Exercise 48




# Exercise 49


Classes
Object properties
Data properties
Class hierarchy: :HighCaloriePizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:FourCheesePizza
:HighCaloriePizza
:InterestingPizza
:LowCaloriePizza
:NamedPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :HighCaloriePizza
Annotations
Description: :HighCaloriePizza
Equivalent To
:Pizza
and (hasCalorificContentValue some xsd:integer[>= 400])
SubClass Of


Class hierarchy: :LowCaloriePizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:FourCheesePizza
:HighCaloriePizza
:InterestingPizza
:LowCaloriePizza
:NamedPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations: :LowCaloriePizza
Annotations
Description: :LowCaloriePizza
Equivalent To
:Pizza
and (hasCalorificContentValue some xsd:integer[< 400])
SubClass Of
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase
hasCalorificContentValue some xsd:integer

# Exercise 50

**Description: :HighCaloriePizza**

Equivalent To

- :Pizza
  and (hasCalorificContentValue some xsd:integer[>= 400])

SubClass Of

- :Pizza

General class axioms

SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase
- hasCalorificContentValue some xsd:integer

Instances

- :ExampleAmericana
- :QuattroFormaggio

**Description: :LowCaloriePizza**

Equivalent To

- :Pizza
  and (hasCalorificContentValue some xsd:integer[< 400])

SubClass Of

- :Pizza

General class axioms

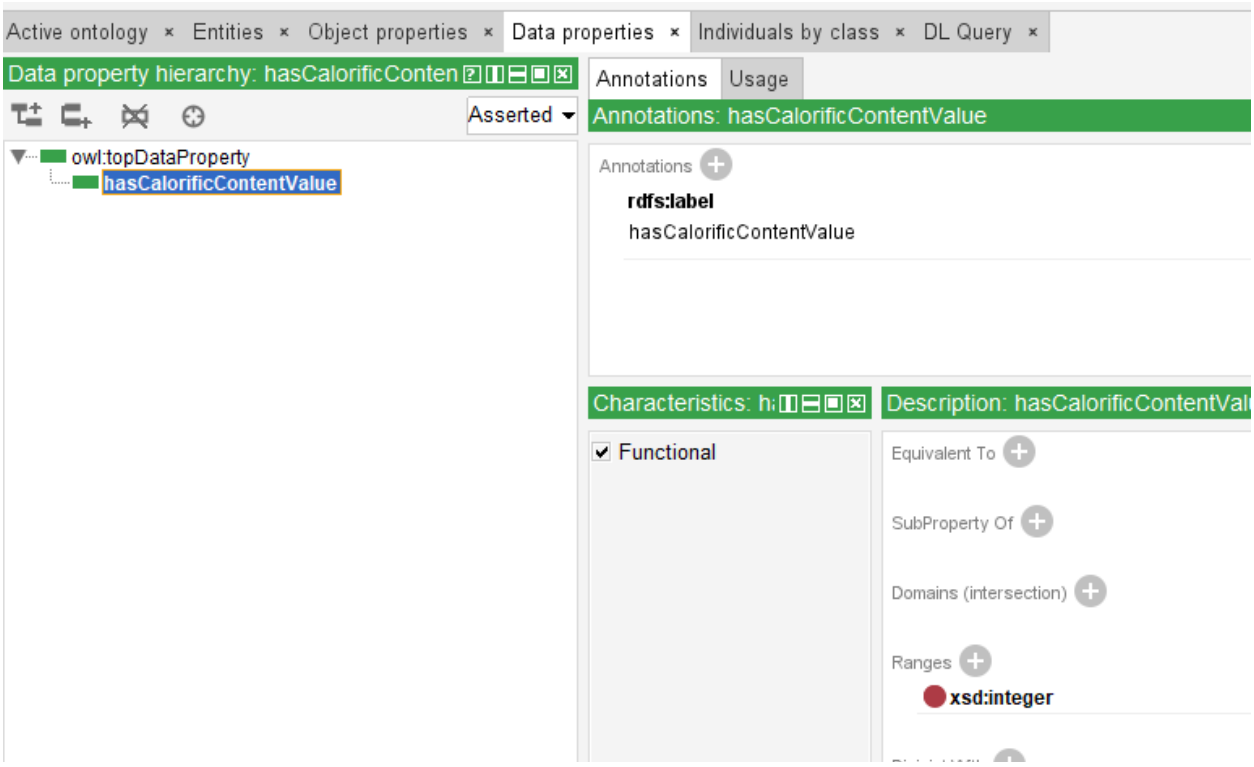
SubClass Of (Anonymous Ancestor)

- hasBase some :PizzaBase
- hasCalorificContentValue some xsd:integer

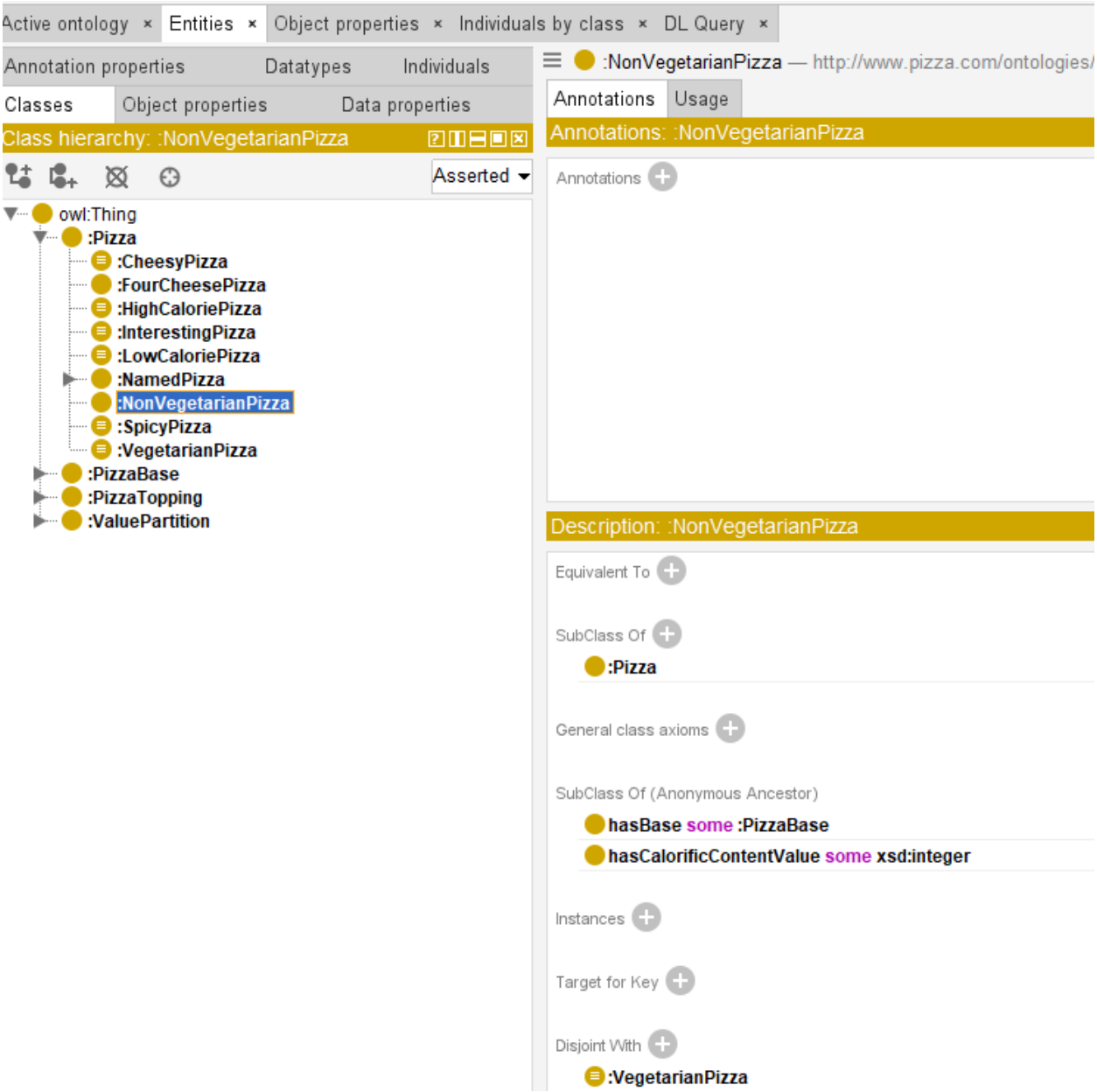
Instances

- :ExampleMargherita
- :ExampleSoho

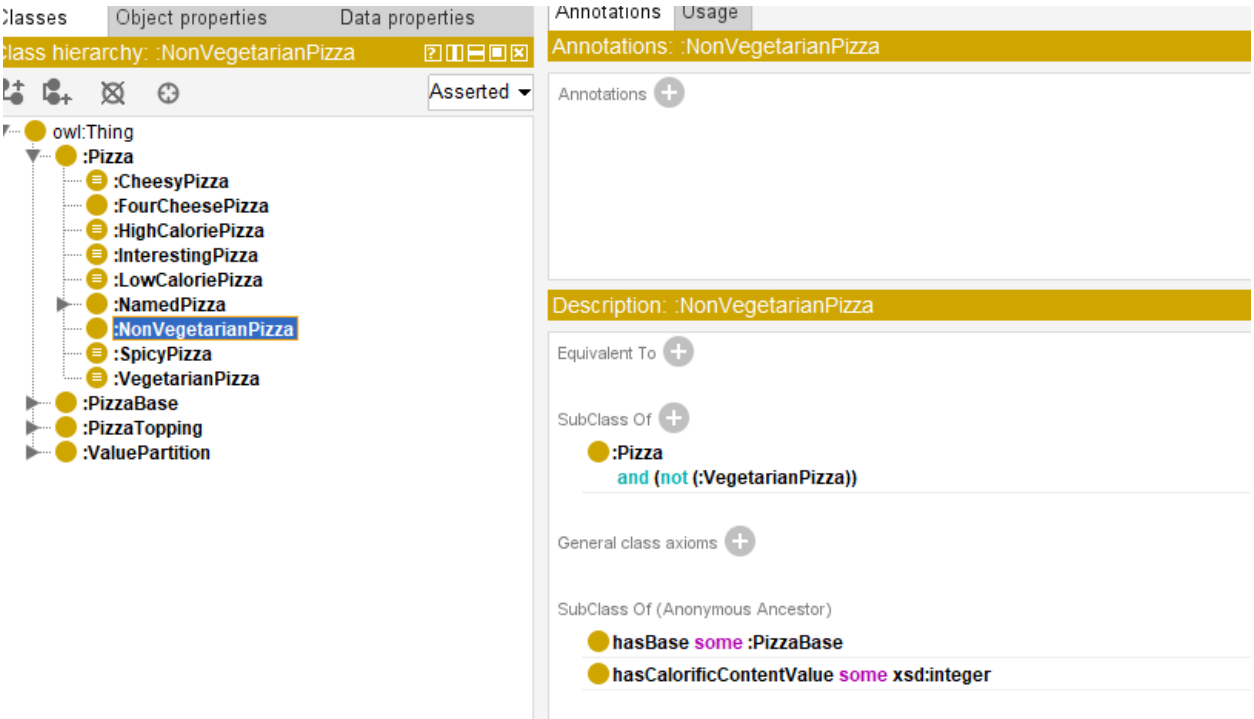
# Exercise 51


Active ontology
Entities
Object properties
Data properties
Individuals by class
DL Query
Data property hierarchy: hasCaloricContent
Asserted
owl:topDataProperty
hasCaloricContentValue
Annotations
Usage
Annotations: hasCaloricContentValue
Annotations
rdfs:label
hasCaloricContentValue
Characteristics: h
Functional
Description: hasCaloricContentVal
Equivalent To
SubProperty Of
Domains (intersection)
Ranges
xsd:integer

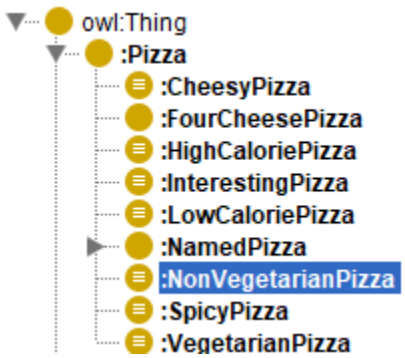
# Exercise 52


Active ontology
Entities
Object properties
Individuals by class
DL Query
Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Class hierarchy: :NonVegetarianPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:FourCheesePizza
:HighCaloriePizza
:InterestingPizza
:LowCaloriePizza
:NamedPizza
:NonVegetarianPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
:NonVegetarianPizza — http://www.pizza.com/ontologies/
Annotations
Usage
Annotations: :NonVegetarianPizza
Annotations
Description: :NonVegetarianPizza
Equivalent To
SubClass Of
:Pizza
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase
hasCalorificContentValue some xsd:integer
Instances
Target for Key
Disjoint With
:VegetarianPizza

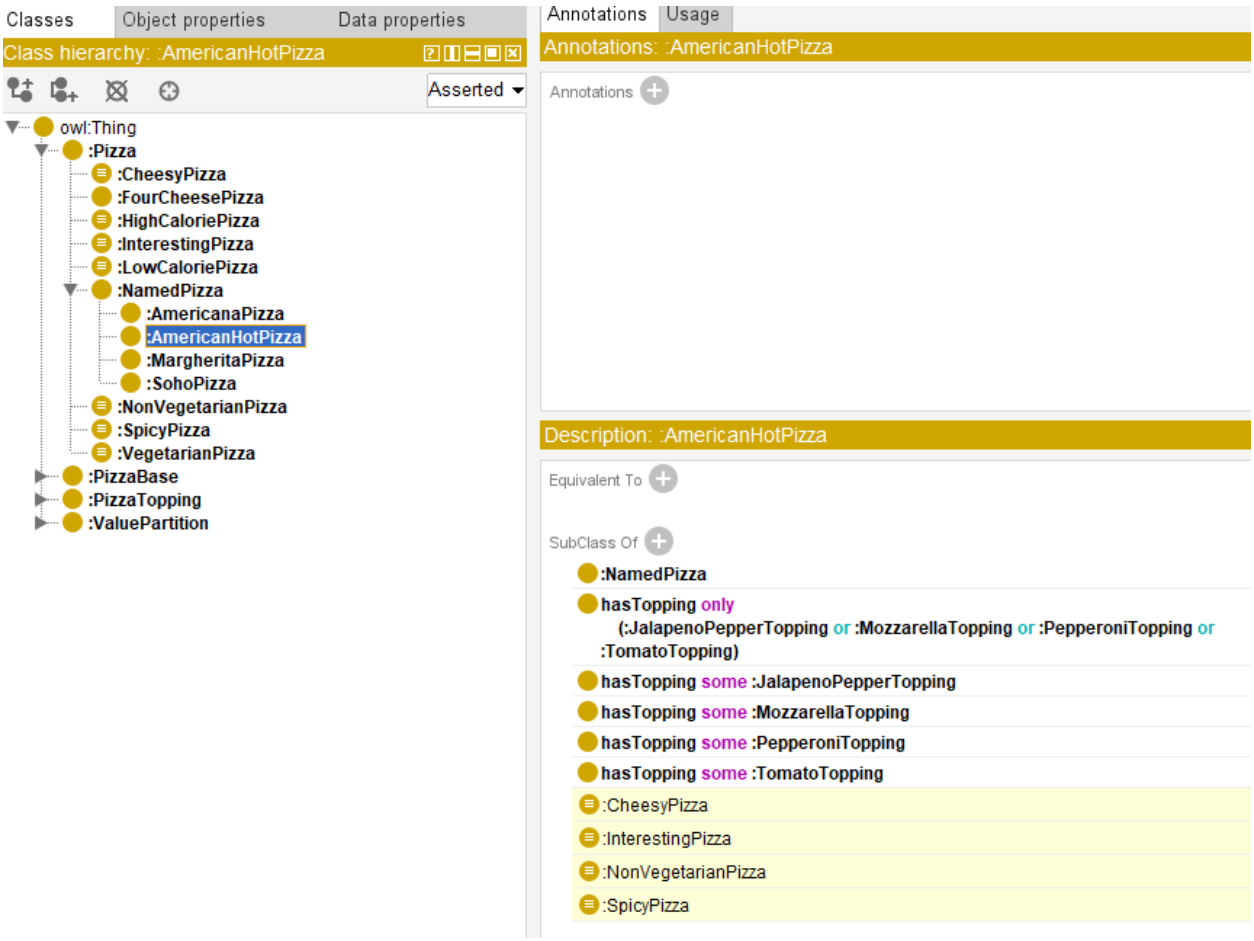
# Exercise 53



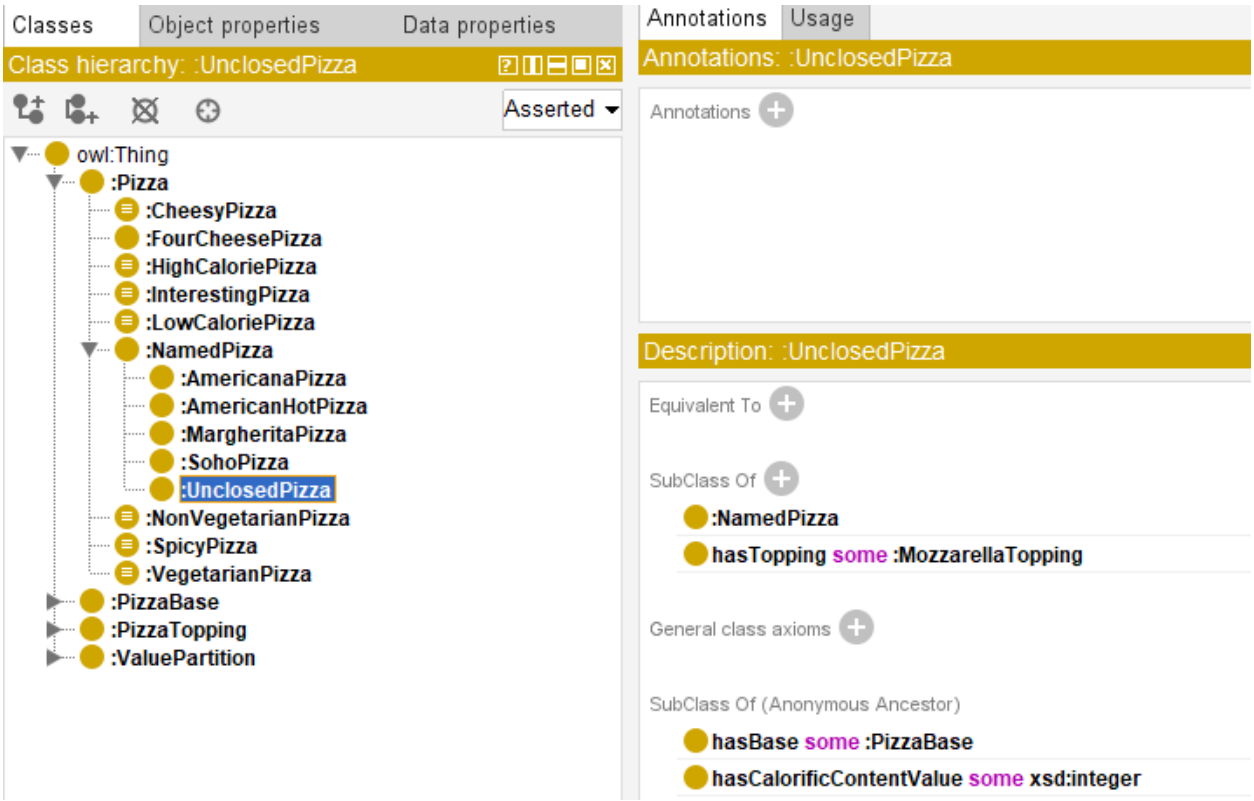

# Exercise 54



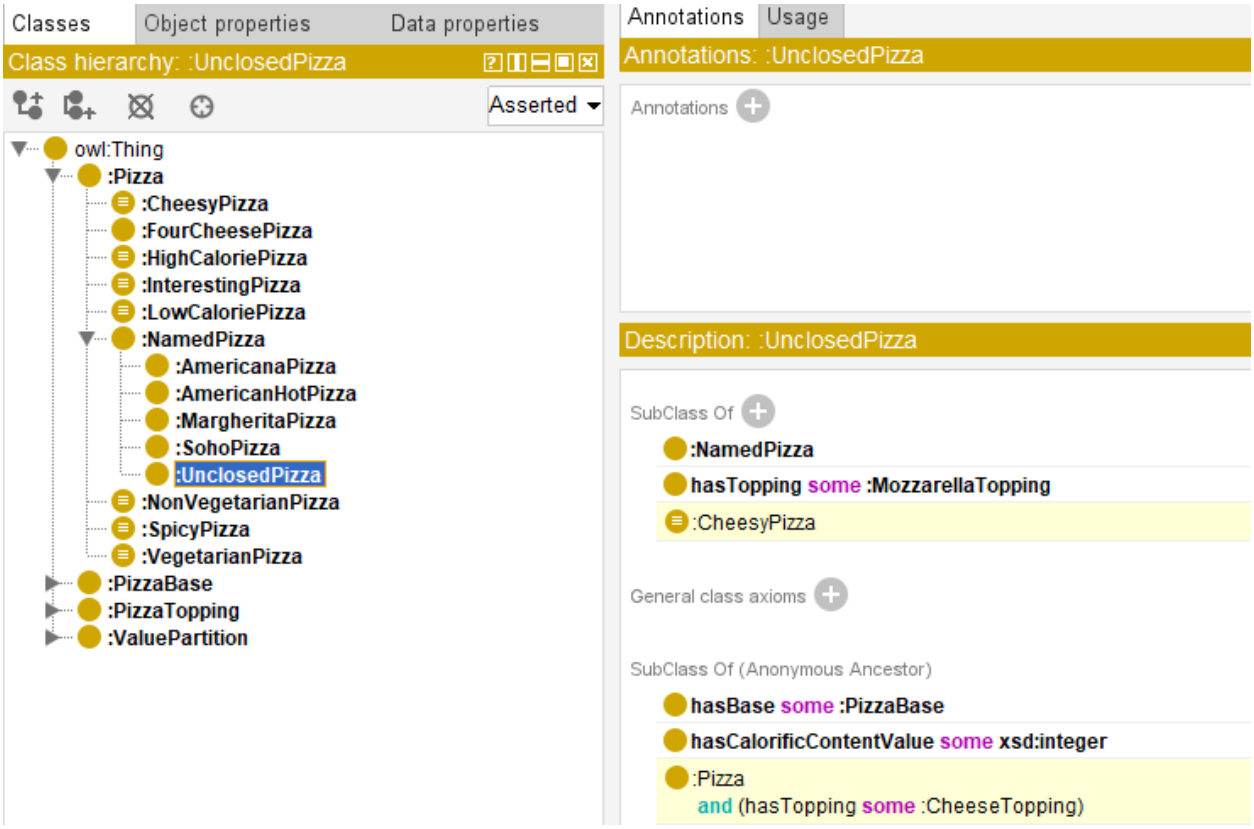

# Exercise 55


Classes
Object properties
Data properties
Class hierarchy: :AmericanHotPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:FourCheesePizza
:HighCaloriePizza
:InterestingPizza
:LowCaloriePizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:NonVegetarianPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :AmericanHotPizza
Annotations
Description: :AmericanHotPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping only
(:JalapenoPepperTopping or :MozzarellaTopping or :PepperoniTopping or :TomatoTopping)
hasTopping some :JalapenoPepperTopping
hasTopping some :MozzarellaTopping
hasTopping some :PepperoniTopping
hasTopping some :TomatoTopping
:CheesyPizza
:InterestingPizza
:NonVegetarianPizza
:SpicyPizza

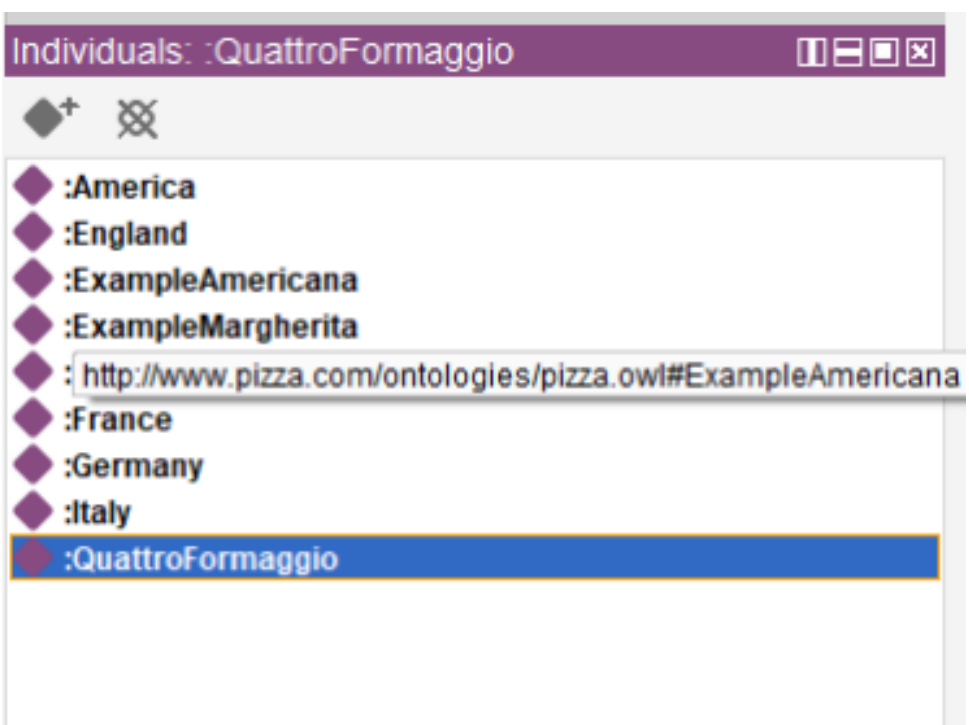
# Exercise 56

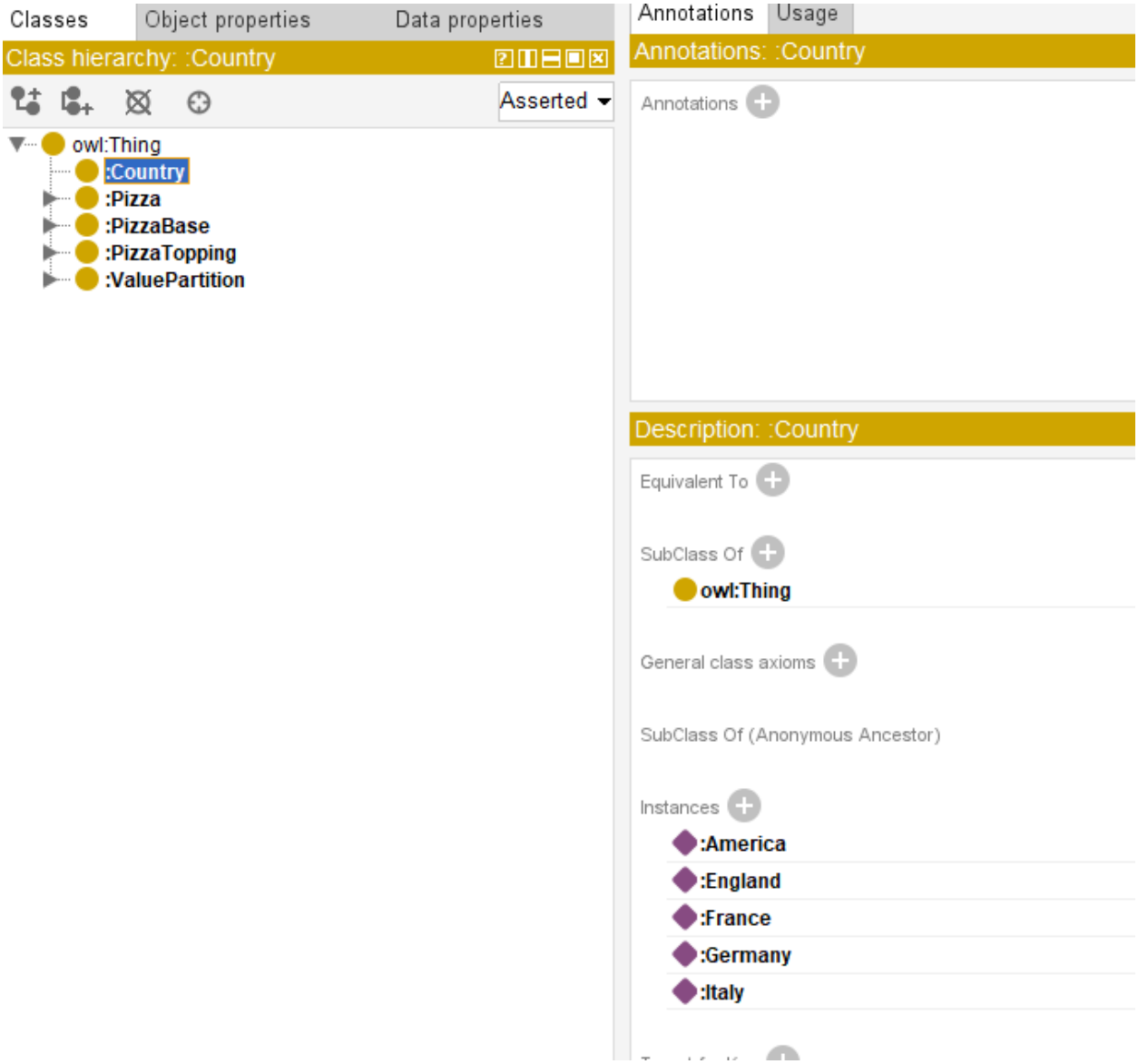

Classes
Object properties
Data properties
Class hierarchy: :UnclosedPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:FourCheesePizza
:HighCaloriePizza
:InterestingPizza
:LowCaloriePizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:UnclosedPizza
:NonVegetarianPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :UnclosedPizza
Annotations
Description: :UnclosedPizza
Equivalent To
SubClass Of
:NamedPizza
hasTopping some :MozzarellaTopping
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase
hasCalorificContentValue some xsd:integer

# Exercise 57

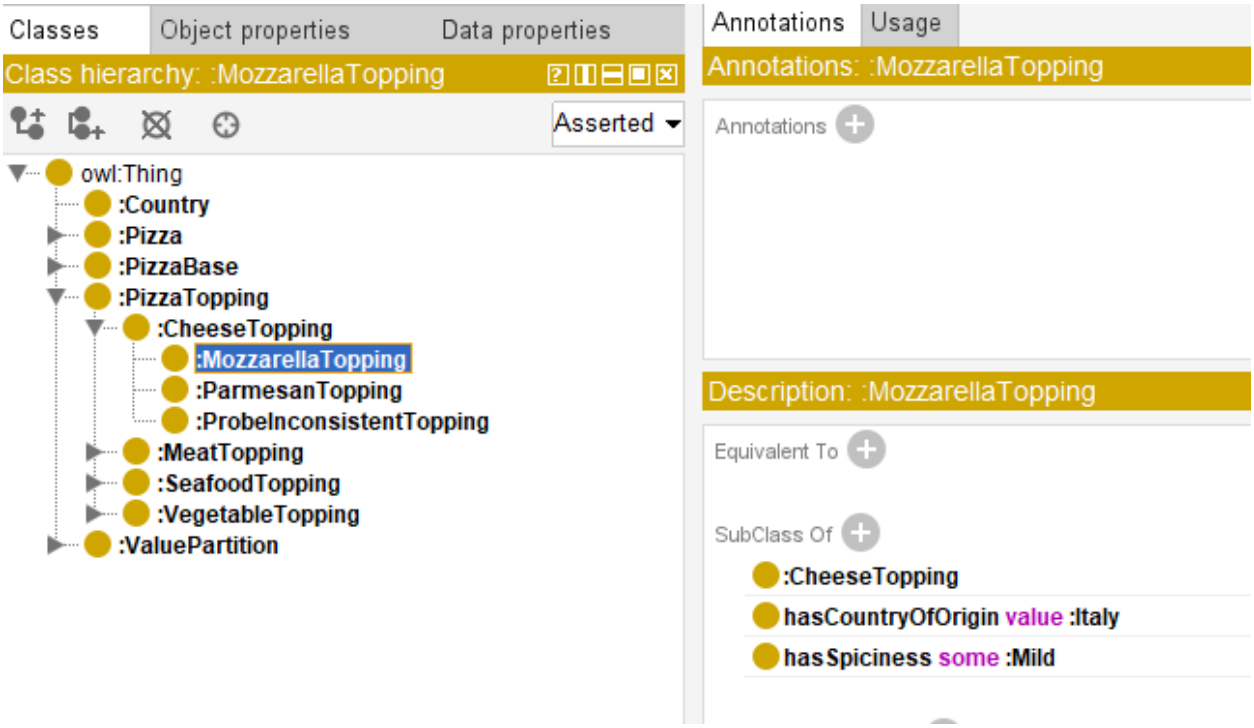

Classes
Object properties
Data properties
Class hierarchy: :UnclosedPizza
Asserted
owl:Thing
:Pizza
:CheesyPizza
:FourCheesePizza
:HighCaloriePizza
:InterestingPizza
:LowCaloriePizza
:NamedPizza
:AmericanaPizza
:AmericanHotPizza
:MargheritaPizza
:SohoPizza
:UnclosedPizza
:NonVegetarianPizza
:SpicyPizza
:VegetarianPizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :UnclosedPizza
Annotations
Description: :UnclosedPizza
SubClass Of
:NamedPizza
hasTopping some :MozzarellaTopping
:CheesyPizza
General class axioms
SubClass Of (Anonymous Ancestor)
hasBase some :PizzaBase
hasCalorificContentValue some xsd:integer
:Pizza
and (hasTopping some :CheeseTopping)

# Exercise 58

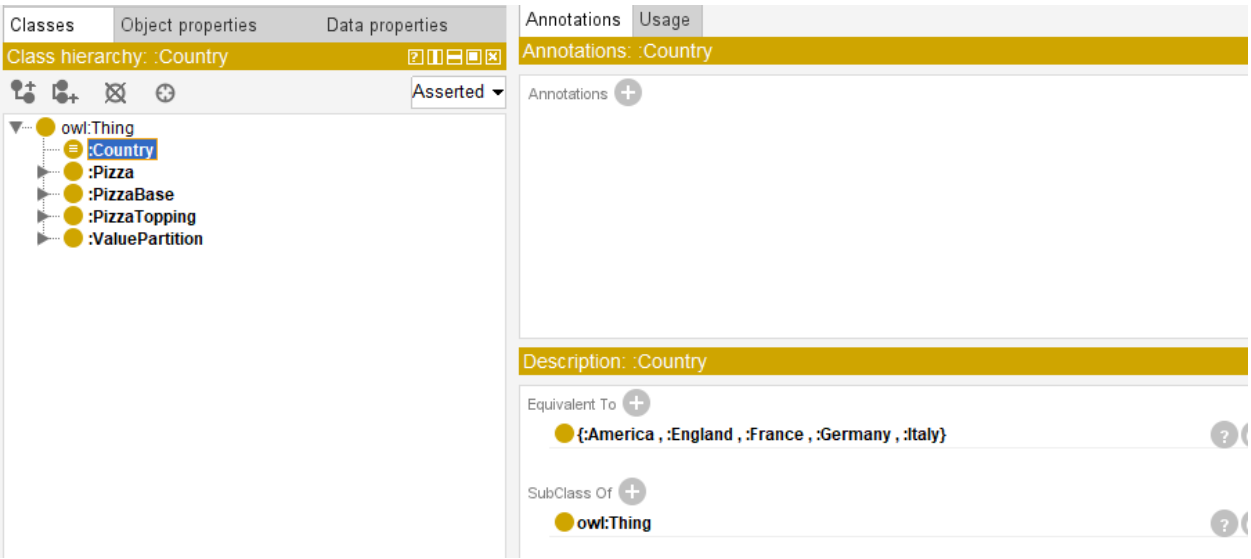

Individuals: :QuattroFormaggio
:America
:England
:ExampleAmericana
:ExampleMargherita
: http://www.pizza.com/ontologies/pizza.owl#ExampleAmericana
:France
:Germany
:Italy
:QuattroFormaggio


Classes
Object properties
Data properties
Class hierarchy: :Country
Asserted
owl:Thing
:Country
:Pizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :Country
Annotations
Description: :Country
Equivalent To
SubClass Of
owl:Thing
General class axioms
SubClass Of (Anonymous Ancestor)
Instances
:America
:England
:France
:Germany
:Italy

# Exercise 59


Classes
Object properties
Data properties
Class hierarchy: :MozzarellaTopping
Asserted
owl:Thing
:Country
:Pizza
:PizzaBase
:PizzaTopping
:CheeseTopping
:MozzarellaTopping
:ParmesanTopping
:ProbeInconsistentTopping
:MeatTopping
:SeafoodTopping
:VegetableTopping
:ValuePartition
Annotations
Usage
Annotations: :MozzarellaTopping
Annotations
Description: :MozzarellaTopping
Equivalent To
SubClass Of
:CheeseTopping
hasCountryOfOrigin value :Italy
hasSpiciness some :Mild

# Exercise 60


Classes
Object properties
Data properties
Class hierarchy: :Country
Asserted
owl:Thing
:Country
:Pizza
:PizzaBase
:PizzaTopping
:ValuePartition
Annotations
Usage
Annotations: :Country
Annotations
Description: :Country
Equivalent To
{:America , :England , :France , :Germany , :Italy}
SubClass Of
owl:Thing